ASPIRE: Agentic /Skills Discovery for Robotics

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https://research.nvidia.com/labs/gear/aspire/

arXiv:2607.00272v1 [cs.RO] 30 Jun 2026

Abstract:
Traditional robot programming is notoriously challenging: it requires orchestrating multimodal perception, managing complex physical contact dynamics, and handling diverse environment configurations and execution failures. We introduce ASPIRE (**A**gentic **S**kill **P**rogramming through **I**terative **R**obot **E**xploration), a continual learning system for robotics that autonomously writes and refines robot control programs in a code-as-policy paradigm while compounding experience into a reusable skill library. ASPIRE enables automated discovery of reusable skills that persist across multiple tasks, simulation and real-world settings, and different embodiments. Rather than relying on fixed, human-engineered pipelines, ASPIRE operates in an open-ended learning loop, consisting of three key components: (1) a closed-loop robot execution engine that exposes fine-grained multimodal traces (e.g., perception overlays, grasp candidates, motion trajectories, and collision feedback), enabling the agent to autonomously diagnose failures, synthesize repairs, and validate outcomes; (2) a continually expanding skill library that distills validated fixes into reusable, transferable robotic knowledge; and (3) an evolutionary search procedure that generates diverse task sequences and control programs, systematically debugging them to explore beyond single-trajectory refinement. As ASPIRE encounters more tasks, its growing skill library enables increasingly rapid adaptation. Consequently, ASPIRE surpasses prior methods by up to 77% on manipulation tasks under perturbation (LIBERO-Pro), 72% on Robosuite's bimanual handover task, and up to 32% on long-horizon household tasks (BEHAVIOR-1K). The accumulated skill library further enables strong zero-shot generalization: on representative unseen long-horizon tasks (LIBERO-Pro Long), ASPIRE achieves 31% success, substantially outperforming the 4% success rate of prior methods despite their heavy reliance on test-time reasoning and retries. Finally, skills discovered in simulation provide initial evidence of sim-to-real transfer, substantially reducing real-robot programming effort despite different embodiments and robot APIs.

1. Introduction

Recent progress in software engineering agents demonstrates that language models can autonomously inspect execution traces, localize failures, revise implementations, and improve through repeated interaction with execution environments [Anthropic, 2025, OpenAI, 2025, anomalyco, 2025, Wang et al., 2025, Yang et al., 2024]. In robotics, this paradigm has inspired *code-as-policy* systems that compose perception modules, planning APIs, and control primitives into executable robot programs [Liang et al., 2023, Singh et al., 2023, Ahn et al., 2022, Huang et al., 2023, Mu et al., 2024, Fu et al., 2026]. Because robot behaviors are represented explicitly as programs, they can in principle be inspected, edited, debugged, and refined through interaction feedback.

However, existing robotic coding agents remain fundamentally limited by naive execution environments that provide only coarse task-level feedback. Debugging robot programs is intrinsically challenging because failures can arise from many interacting components, including multimodal perception, motion planning, grasp generation, contact dynamics, and long-horizon task coordination. A failed rollout may indicate that the task did not succeed, but not whether the root cause is incorrect perception, an unstable grasp, a planning error, or a downstream recovery failure. Without

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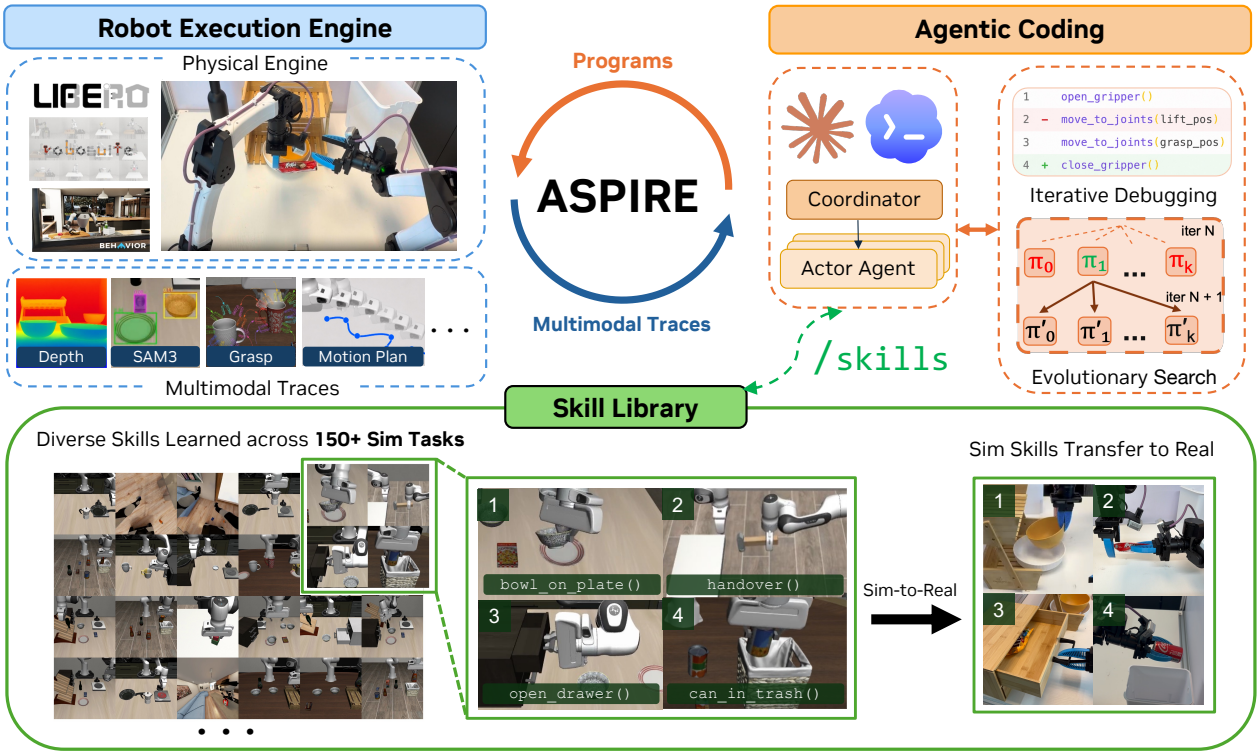


Figure 1: **Aspire system overview.** A *coordinator* spawns an *actor agent* (coding agent) per task, enabling parallel learning across tasks. Each actor refines and validates robot programs through *iterative debugging* with the *robot execution engine*, which exposes per-primitive multimodal traces for failure attribution and repair. *Evolutionary search* samples diverse candidate programs $(\pi_0, \ldots, \pi_k)$, sends each through the engine, and conditions the next generation $(\pi'_0, \ldots, \pi'_k)$ on surviving programs and residual failure traces. The *coordinator* writes validated repairs into a shared *skill library*, which future actors retrieve as in-context guidance; skills discovered in sim can also be adapted as cross-embodiment guidance for real-robot programming.

fine-grained diagnostic traces, agents have limited ability to determine what evidence to inspect, how to localize failures, or what repair strategy to attempt.

Moreover, existing systems do not accumulate experience across tasks. Once a task is completed, discovered fixes and recovery strategies are discarded rather than consolidated into reusable skills. As a result, the agent solving its hundredth task is effectively no more experienced than the agent solving its first.

Human robotics engineers take a fundamentally different approach. When a robot program fails, they replay executions, inspect perception outputs and motion trajectories, localize the failing subsystem, revise the implementation, and internalize reusable recovery strategies. Over time, debugging experience compounds into transferable knowledge, including grasp recovery heuristics, navigation strategies, prompting recipes, and procedural fixes that generalize across tasks. This accumulation of reusable knowledge is a key reason why human robot programmers become progressively more effective over time.

In this work, we introduce ASPIRE, a self-improving continual learning robotic system that autonomously writes and refines robot control programs in a code-as-policy paradigm while accumulating experience into a reusable skill library. Rather than operating within a fixed perception-plan-execute

pipeline, ASPIRE runs in an open-ended learning loop, in which the agent self-determines how to inspect execution traces, diagnose failures, synthesize repairs, validate corrected behaviors, and consolidate successful recovery patterns into persistent skills that transfer across tasks.

An overview of ASPIRE is shown in Fig. 1. ASPIRE is built from three components. First, a **closed-loop robot execution engine** replaces coarse rollout-level feedback with per-primitive execution traces. For each perception, planning, grasping, and control call, the execution engine records the observations, inputs, outputs, and visual evidence if possible. These rich multimodal traces allow the agent to selectively inspect salient primitive logs, progressively localize failures, and validate repairs through re-execution. Second, ASPIRE maintains a growing **skill library** that distills validated fixes into reusable, transferable robotic knowledge retrievable as in-context guidance for future tasks. Third, ASPIRE employs an **evolutionary search procedure** that generates diverse task sequences and control programs, exploring beyond single-trajectory self-improvement through iterative debugging and parallel refinement. Together, these components establish a self-improving robotic system whose performance scales with experience: the more tasks ASPIRE sees, the larger its skill library grows, and the more it transfers to novel tasks, longer-horizon behaviors, and real-world robotic settings where similar failure-recovery patterns emerge under different embodiments.

Empirically, ASPIRE demonstrates strong self-improvement across diverse short- and long-horizon robot benchmarks. Against prior coding agents [Fu et al., 2026], ASPIRE improves success rate by up to 77 points on LIBERO-Pro [Zhou et al., 2025] perturbation suites, by up to 72 points on Robosuite [Zhu et al., 2020] contact-rich manipulation tasks, and by up to 32 points on BEHAVIOR-1K [Li et al., 2024] long-horizon household tasks with procedurally generated layouts. Based on the skill library accumulated on LIBERO-90 [Liu et al., 2023], ASPIRE transfers zero-shot to LIBERO-Pro Long [Zhou et al., 2025], and reaches 31% success, while prior methods saturate at 4% despite their reliance on test-time reasoning and retries. We further evaluate three simulation-discovered skills on a real bimanual robot, showing that retrieving skills discovered in sim as in-context guidance reduces real-world reasoning tokens and enables successful programs where naive debugging without skill transfer fails entirely.

2. Method

ASPIRE consists of three components that together form an open-ended learning loop: (1) a *robot execution engine* (§2.1) that exposes per-primitive multimodal traces for failure attribution and executes agent-written repairs for closed-loop validation; (2) a *skill library* (§2.2) that accumulates validated repairs as reusable skills; and (3) an *evolutionary search* procedure (§2.3) that broadens exploration beyond single-trajectory repair. As ASPIRE encounters more tasks, its skill library grows, allowing future tasks to inherit accumulated repairs and reusable strategies.

As shown in Fig. 1, ASPIRE adopts a coordinator–actor architecture. A central coordinator manages the shared skill library and dispatches actor coding agents to individual tasks, while each actor writes, executes, diagnoses, and repairs robot programs within the robot execution engine. Actors do not exchange full chat histories or raw rollout trajectories. Instead, transferable experience is distilled into the skill library, allowing each actor's context window to remain focused on the task specification, current program, and structured execution traces associated with the current failure.

2.1. Robot Execution Engine

Embodied coding agents need execution evidence to debug robot programs. Prior methods expose this evidence through fixed, human-designed interfaces, typically as manually curated scene-level summaries or a set of pre-defined observations. This creates a trade-off: too little evidence can hide

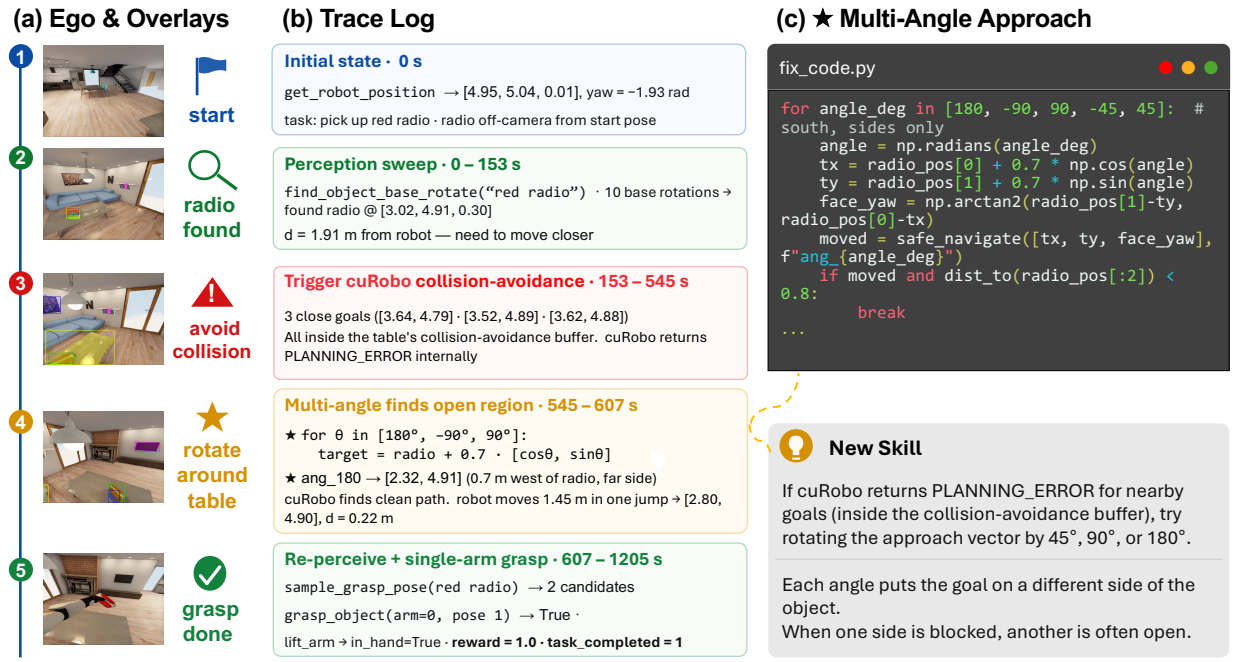


Figure 2: **Robot execution engine.** Trace-guided debugging on a BEHAVIOR-1K NAVIGATE-AND-PICK-UP-RADIO task. (a) Ego-view keyframes and overlays show the robot locating the radio but failing to approach it. (b) The primitive trace localizes the failure to repeated `PLANNING_ERROR`s: candidate navigation goals fall inside the table's collision-avoidance buffer. (c) The agent patches the program with a multi-angle approach routine, re-perceives the radio from a reachable side, and completes the grasp. The validated repair is admitted as a reusable *Multi-Angle Approach* skill.

the failing primitive, while too much raw visual context can distract the agent from the causal chain that produced the failure.

ASPIRE turns this fixed feedback channel into an open-ended debugging environment. The robot execution engine records per-primitive multimodal traces for perception, planning, and control calls, exposes the trace to the coding agent, and executes agent-written repairs for closed-loop validation. For each primitive call, the trace stores the invoked API, inputs and outputs, return status, and relevant multimodal evidence such as RGB keyframes, overlays, grasp candidates, object poses, and motion-planning results. The agent does not receive full video frames; the engine keeps frames immediately before and after each primitive call together with the corresponding overlays and return values, so the agent can focus on evidence around calls implicated by the failure.

Fig. 2 zooms in on one BEHAVIOR-1K NAVIGATE-AND-PICK-UP-RADIO debugging episode. The ego-view keyframes show that the robot finds the radio, repeatedly fails to approach it, and then succeeds after changing its approach direction. The primitive trace localizes the failure: perception succeeds and returns a radio pose, but repeated `navigate_to_pose` calls return `PLANNING_ERROR`. By checking the navigation return values and associated logs, the agent finds that the generated navigation target lies too close to the table boundary, approximately within 20 centimeters of the table edge. This triggers collision avoidance and causes the planner to fail. Thus, the failure is not due to detecting or grasping the radio, but rather to the infeasibility of the target pose under the table's collision constraints.



Figure 3: **Skill library.** ASPIRE stores validated, agent-discovered repair knowledge as reusable in-context skills rather than a fixed set of human-written primitives. Top: representative entries show learned skills about localization disambiguation, motion-primitive construction, and navigation recovery. Middle: the library grows across heterogeneous categories, including localization, navigation, motion primitives, object-level grasping, scene understanding, and debugging workflows. Bottom: selected skills discovered in sim are used as in-context guidance for real-robot programming, providing evidence that skills can transfer across embodiments. Additional skill examples and prompts are shown in Appendix A.

The repair follows directly from this diagnosis. Rather than changing the perception prompt or grasp primitive, the agent writes a multi-angle approach routine that samples alternative navigation targets around the radio and executes an approach direction that clears the collision buffer. The execution engine exposes the evidence, validates the patched program, and enables the agent to analyze the resulting logs, form a hypothesis, and make a targeted repair decision.

2.2. Skill Library

Program failures recur across tasks, but the reusable knowledge is rarely an entire task program. ASPIRE's skill library stores heterogeneous repair knowledge: localization heuristics, perception prompts, grasping constraints, navigation recovery strategies, motion primitives, scene-understanding routines, and debugging workflows. We do not prescribe this taxonomy in advance. Skills are induced from validated repairs: the coding agent diagnoses a failure from execution traces, patches the program, validates the fix on debugging configurations, and the coordinator admits only reusable patterns into the shared library.

Each skill is stored as compact in-context guidance, including the failure signature, when-to-apply condition, repair strategy, and, when useful, a representative code sketch. Fig. 3 shows representative entries and the resulting library breadth. For the radio task in Fig. 2, the admitted skill is a navigation recovery pattern rather than a complete radio-pickup program: when planner errors recur near an obstacle boundary because sampled target poses fall inside the collision buffer, sample alternative approach directions around the object before retrying perception and grasping. This representation lets future actors reuse validated repairs instead of rediscovering them through test-time reasoning, supports zero-shot transfer to harder simulated tasks, and provides the mechanism for selected simulation-discovered skills to generalize across embodiments and transfer to real robots.

Actors report structured findings that summarize the failure mode, validated fix, and potentially transferable repair pattern. The coordinator audits these findings, verifies compliance with the allowed API policy, and promotes only reusable repairs that have passed debug validation into the shared skill library. Appendix E.1 provides the prompts used for actor reporting and coordinator-guided skill admission.

2.3. Evolutionary Search

Trace-guided debugging alone can collapse into local repair loops, where the agent repeatedly patches the same failed strategy instead of exploring fundamentally different ways to solve the task. ASPIRE uses evolutionary search to broaden exploration of executable robot programs, encouraging diverse repair hypotheses and task strategies.

In each round, based on the skill library, the coding agent proposes a population of K candidate programs conditioned on the top-performing previous programs and failure traces from previous evaluations. Each candidate is executed in the robot execution engine, producing task outcomes together with new diagnostic traces. The next round is then conditioned on the best-performing programs together with their remaining failure modes, allowing the search to explore distinct strategies rather than repeatedly refining the same solution.

The search target is the robot program itself. Candidates are selected through closed-loop execution, and validated repairs are admitted into the skill library after search concludes, provided they generalize across environment variations and tasks. Search terminates when a candidate solves the debugging configurations or when the search budget is exhausted. Algorithm 1 summarizes the evolutionary search procedure. See Appendix E.4 for pipeline details.

Algorithm 1 Evolutionary search over programs

Require: task τ, program P^0, sets $S_{\text{dbg}}, S_{\text{val}}$, skill library $\mathcal{L}$, agent M, budget (K, T), threshold θ
Notation: $\textsc{Execute}(P, S) = (r, Z)$ returns score r and trace bundle Z.
$(r^\star, Z^0) \leftarrow \textsc{Execute}(P^0, S_{\text{dbg}})$; $P^\star \leftarrow P^0$
$\mathcal{H} \leftarrow \{(P^0, r^\star, Z^0)\}$
for $i = 1, \ldots, T$ **do**
  $\{P_i^k\}_{k=1}^K \leftarrow \textsc{ProposeRepairs}(M, \tau, \text{Top3}(\mathcal{H}), \mathcal{L}, \mathcal{H})$
  for $k = 1, \ldots, K$ **do**
    $(r_i^k, Z_i^k) \leftarrow \textsc{Execute}(P_i^k, S_{\text{dbg}})$
  end for
  $\mathcal{H} \leftarrow \mathcal{H} \cup \{(P_i^k, r_i^k, Z_i^k)\}_{k=1}^K$; $k^\star \leftarrow \arg\max_k r_i^k$
  if $r_i^{k^\star} > r^\star$ **then**
    $(P^\star, r^\star) \leftarrow (P_i^{k^\star}, r_i^{k^\star})$
  end if
  if $r^\star \geq \theta$ **then**
    break
  end if
end for
$(r_{\text{val}}, Z_{\text{val}}) \leftarrow \textsc{Execute}(P^\star, S_{\text{val}})$
$\mathcal{G} \leftarrow \textsc{ExtractValidatedPatterns}(\mathcal{H}, P^\star, r_{\text{val}}, Z_{\text{val}})$
return $(P^\star, r_{\text{val}}, \mathcal{G})$

3. Experiments

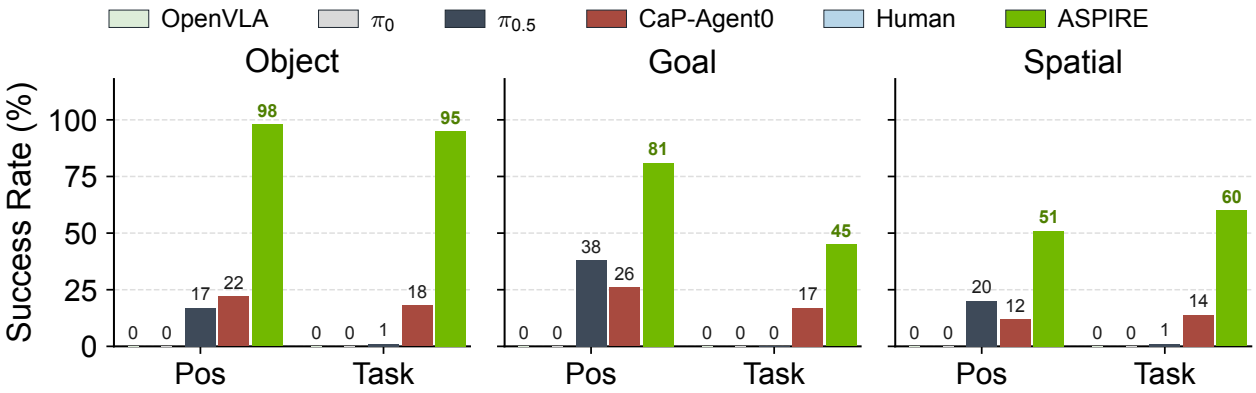
3.1. Experimental Setup

For all the simulation benchmarks, we use Claude Code [Anthropic, 2025] with Claude Opus 4.6 and a 1M-token context window as the coding agent. The agent writes executable Python robot programs in CaP-X [Fu et al., 2026], an open-source code-as-policy framework built on MuJoCo Playground [Zakka et al., 2025], with robot programming APIs for perception, geometry, and motion planning. The agent, environment, and API set are fixed across all experiments.

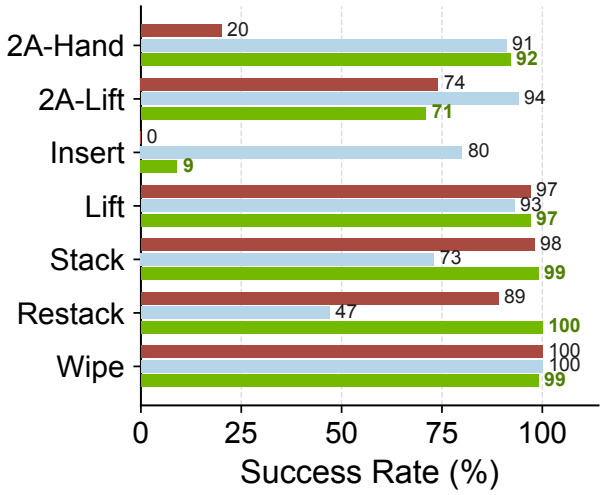
For the real-robot skill transfer study (§3.6), we use OpenAI Codex GPT-5.5 in reasoning-xhigh mode on a bimanual YAM manipulation station. We select three skills compiled by ASPIRE in Franka-based simulation: soda-can pickup, bowl-on-plate placement, and drawer push/pull. We provide these as in-context guidance to the real-robot coding agent. The real-robot setting uses a different embodiment and API from simulation, but exposes ASPIRE multimodal execution traces so the agent can autonomously run, inspect, and debug programs without task-specific human guidance during the debugging loop. We compare runs with and without the corresponding simulation-discovered skill, measuring tokens to first success and held-out real-robot success rate.

3.2. Benchmarks and Baselines

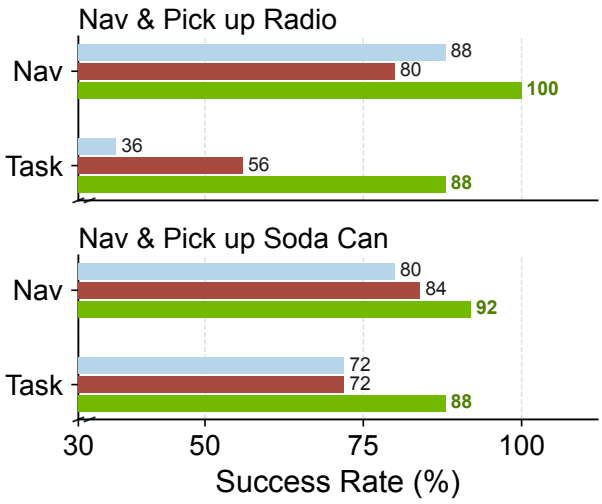
We evaluate ASPIRE on three benchmark families: LIBERO-Pro [Zhou et al., 2025] for short-horizon robustness under object, goal, and spatial perturbations; Robosuite [Zhu et al., 2020] for contact-rich single- and dual-arm manipulation; and BEHAVIOR-1K [Li et al., 2024] for long-horizon household mobile manipulation on NAVIGATE-AND-PICK-UP-SODA-CAN and NAVIGATE-AND-PICK-UP-RADIO. Our primary coding-agent baseline is CaP-Agent0 [Fu et al., 2026], which uses visual differencing, a predefined skill library, and per-episode test-time retries. We also compare with end-to-end VLA policies, including OpenVLA [Kim et al., 2024], π_0 [Black et al., 2025], and $\pi_{0.5}$ [Physical Intelligence et al., 2025]. For zero-shot transfer, we evaluate held-out LIBERO-Pro Long tasks using the skill library accumulated on LIBERO-90 [Liu et al., 2023].



(a) **LIBERO-Pro** [Zhou et al., 2025]: macro-averaged success over 10 tasks × 50 held-out environment seeds per suite/perturbation. For each task, ASPIRE learns and collects skills on seeds 51–65 and evaluates one generated program on seeds 1–50.



(b) **Robosuite** [Zhu et al., 2020]: success over 100 held-out trials per task. ASPIRE learns on seeds 101–125 and evaluates one generated program per task on seeds 1–100; 2A tasks are bimanual.

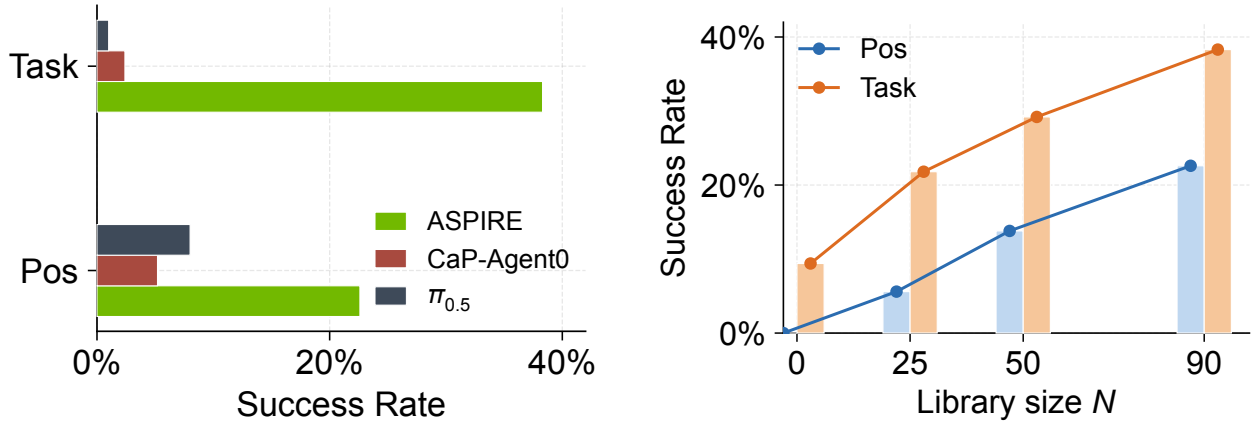


(c) **BEHAVIOR-1K** [Li et al., 2024]: long-horizon mobile manipulation on two household tasks. ASPIRE learns on seeds 26–35 and evaluates on seeds 1–25 with incremental block execution; Navigation and Task success are reported separately.

Figure 4: **Aspire improves over prior coding agents and end-to-end VLAs across three benchmark families.** (a) Short-horizon manipulation on LIBERO-Pro; (b) contact-rich manipulation on Robosuite; (c) long-horizon mobile manipulation on BEHAVIOR-1K. ASPIRE evaluates one generated program per task across held-out seeds, while CaP-Agent0 regenerates a separate program per seed with test-time reasoning and retries. ASPIRE improves performance across all benchmarks, with several results surpassing programs written by human experts. Detailed per-task results are in Appendix B and D.1.

3.3. Evaluation Protocol

Across all benchmarks, an environment seed fixes each task instance, including object poses, distractors, and initial robot/object states. We use disjoint debug and evaluation seeds: ASPIRE learns on a small debug split, then reports success on larger held-out evaluation seeds with one generated program per LIBERO-Pro/Robosuite task, while CaP-Agent0 regenerates a separate program for each seed with test-time reasoning and retries. For BEHAVIOR-1K evaluation, ASPIRE uses incremental block execution, generating each next code block from the current multimodal



(a) Full N=90 library zero-shot results versus baselines.

(b) Zero-shot success improves as skill-library size increases.

Figure 5: **Cross-task zero-shot transfer on LIBERO-Pro Long.** Skills accumulated on LIBERO-90 improve zero-shot performance on held-out long-horizon tasks. Figure (a) compares the full N=90 library with baselines. Figure (b) shows Pos/Task success as the size of the skill library increases. All success rates are macro-averaged over 10 tasks per axis. Per-task results are in Appendix C.

trace.

3.4. Main Evaluation Results

Figure 4 summarizes the main evaluation. "Human" means the programs are written by human experts. On LIBERO-Pro, ASPIRE improves over all the baselines on all three suites: averaging the Pos and Task perturbation axes, ASPIRE gains 77% on Object, 41.5% on Goal, and 42.5% on Spatial over the strongest baseline in each suite. $\pi_{0.5}$ is stronger than OpenVLA and π_0 on some position perturbations, but remains far below ASPIRE and largely collapses under task paraphrases. On Robosuite, ASPIRE preserves near-saturated performance on easier contact-rich tasks and substantially improves bimanual handover, from 20% to 92%. On BEHAVIOR-1K, ASPIRE outperforms both human and CaP-Agent0 on navigation and task success, with the largest task-level gain over CaP-Agent0 on NAVIGATE-AND-PICK-UP-RADIO, from 56% to 88%.

3.5. Zero-Shot Transfer to Unseen Tasks

We evaluate whether repair skills accumulated on LIBERO-90 transfer zero-shot to held-out LIBERO-Pro Long tasks. ASPIRE builds library snapshots from $N \in \{0, 25, 50, 90\}$ source tasks, where N=0 is an empty-library setting and N=90 is the full suite. For each held-out task, ASPIRE generates one program and evaluates it across seeds with no additional debugging, retries, or task-specific library updates.

Figure 5(a) compares the full N=90 skill library with prior baselines on the two LIBERO-Pro Long transfer axes: ASPIRE reaches 23% success on position perturbations and 38% on task perturbations, outperforming CaP-Agent0 and $\pi_{0.5}$ on both axes. Figure 5(b) shows that success increases consistently as the size of the skill library grows, indicating that validated repairs from short-horizon tasks provide reusable robotic knowledge for longer-horizon compositions.

Table 1: **Real-robot cross-embodiment skill transfer.** For each task, we compare real-robot debugging with and without retrieving a corresponding simulation-discovered skill from ASPIRE's library. Token counts are measured until the first successful real-robot program, or until the debugging budget is exhausted for runs that do not reach success. Success rate reports evaluation trials for the generated programs.

Task	Output Tokens (M) ↓		Total Tokens (M) ↓		Success Rate ↑	
	w/o Skills	w/ Skills	w/o Skills	w/ Skills	w/o Skills	w/ Skills
Put bowl on plate	0.05	0.04	8.65	5.11	20/20	20/20
Lift soda can	0.18	0.03	61.94	6.58	13/20	19/20
Open/push drawer	1.33	0.36	334.917	81.67	0/20	11/20

3.6. Real-Robot Skill Transfer Across Embodiments

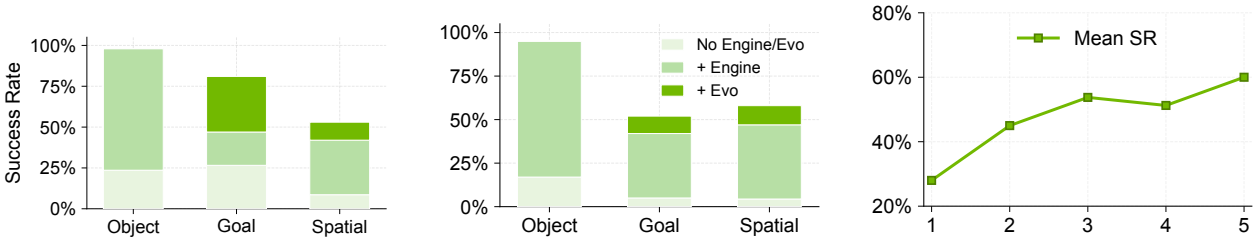
We evaluate whether skills learned in simulation can reduce debugging effort on a real robot with a different embodiment. The transfer is not a direct policy deployment: the real robot uses its own perception, calibration, and control stack, and the coding agent must still adapt programs through real-world execution feedback. The question is whether retrieving simulation-discovered skills provides useful in-context guidance that reduces the amount of real-world debugging needed to reach a successful program.

Table 1 shows that transferred skills consistently reduce debugging cost, while their effect on final success is task-dependent. Bowl placement succeeds in both settings but uses fewer tokens with skill retrieval; soda-can lifting improves from 13/20 to 19/20 while reducing total tokens by nearly an order of magnitude; and drawer manipulation reaches 11/20 success with skill guidance, while the no-skill baseline exhausts a larger token budget without producing a successful evaluation program. These results indicate that selected failure-derived skills can guide real-robot program synthesis across embodiment and API changes rather than merely memorizing simulator-specific code.

3.7. Ablation Studies

Robot execution engine and evolutionary search. We ablate two key components of ASPIRE on LIBERO-Pro: the robot execution engine, which exposes dense execution traces and validates task-level repairs, and evolutionary search, which explores additional repair candidates after robot-execution-engine debugging. Figure 6(a) and Figure 6(b) decompose success into a base system without either component, the gain from adding the robot execution engine, and the additional gain from evolutionary search. The robot execution engine provides the largest average improvement, raising macro-average success from 14% to 62%; evolutionary search further improves the remaining hard tasks, reaching 72% with both components.

Evolutionary search iterations. Figure 6(c) tracks performance on low-performing tasks as the evolutionary-search budget increases. Success improves steadily in the first few rounds, suggesting that sampling multiple repair hypotheses quickly recovers alternatives missed by single-iteration debugging. The curve continues to increase more gradually afterward, indicating that additional rounds still help on residual hard cases, but with diminishing returns.



(a) Pos perturbation. (b) Task perturbation. (c) Per-iteration progress.

Figure 6: **Robot execution engine and evolutionary search ablations on LIBERO-Pro.** Figures (a) and (b) show stacked bars for position and task perturbations: the base system without the robot execution engine or evolutionary search, the gain from adding the robot execution engine, and the additional gain from evolutionary search. On average, the robot execution engine provides the largest gain, raising macro-average success from 14% to 62%; evolutionary search further improves remaining hard tasks. Figure (c) shows average progress over low-performing tasks across evolutionary-search iterations. Per-task results are in Appendix D.

4. Related Work

Agentic robot control. Robot control has been studied through end-to-end vision-language-action policies and executable programs that compose perception, planning, and control APIs [Ahn et al., 2022, Brohan et al., 2023, Octo Model Team et al., 2024, Kim et al., 2024, Black et al., 2025, Mu et al., 2024, Physical Intelligence et al., 2025, NVIDIA et al., 2025, Sun et al., 2026, Li et al., 2026, Fu et al., 2026]. Software-engineering agents provide a related write-execute-debug loop for code [Jimenez et al., 2024, Wang et al., 2025, Yang et al., 2024, Wang et al., 2024a, Chen et al., 2024, Anthropic, 2025, OpenAI, 2025, anomalyco, 2025]. ASPIRE builds on the executable-program paradigm, but focuses on persistent embodied improvement: the agent freely selects primitive-level multimodal traces, writes repairs, validates, and preserves successful experiences across tasks.

Self-improving agents and skill libraries. LLM agents have been improved through open-ended memory, skill libraries, and self-evolving skill repositories [Wang et al., 2023, Tziafas and Kasaei, 2024, Peng et al., 2024, Fu et al., 2025, Zhang et al., 2026, Xie et al., 2026]. Other systems use LLMs to generate rewards, curricula, environments, or search candidates [Ma et al., 2024a, Yu et al., 2023, Xie et al., 2024, Liang et al., 2024, Ma et al., 2024b, Du et al., 2023, Wang et al., 2024b, Romera-Paredes et al., 2024, Novikov et al., 2025, Guo et al., 2025, 2026, Cao et al., 2026]. Unlike success-only memories, textual reflections, or reward functions for downstream policy training, ASPIRE stores validated repair knowledge extracted from attributed embodied failures. Its skills are open-ended in both content and admission: they emerge from the agent's own debugging experience, span heterogeneous categories (§2.2), and are reused as in-context guidance for future program repair. Evolutionary search (§2.3) further broadens the space of executable program repairs before reusable patterns are admitted to the library.

5. Limitations

ASPIRE has several important limitations. First, while selected simulation-discovered skills support autonomous real-robot debugging on a different embodiment, our system is not yet a fully autonomous real-world lifelong learner. Unlike simulation, where success checking and scene resets are cheap and programmatic, real-world deployment still requires robust success detection, safe reset, safety monitoring, and calibration maintenance. Future work should close this evaluation-and-reset loop to

scale sim-to-real skill transfer across broader real-world task suites. Second, our method relies on a frozen frontier LLM (Claude Opus 4.6 with a 1M-token context window) to interpret multimodal traces, write program repairs, and propose evolutionary-search candidates; we have not verified that smaller or weaker LLMs can sustain the same debugging loop. Third, ASPIRE writes programs using a predefined API of perception, planning, and control primitives. This API makes debugging tractable and safe, but also bounds the behaviors the agent can express: if a task requires sensing, control, or interaction capabilities outside the exposed primitives, the agent must either approximate them inefficiently or rely on humans to extend the API. Future systems should study how agents can safely propose, validate, and incorporate new robot primitives. Fourth, the skill library currently prioritizes validated reusable repairs but does not fully solve long-term memory management. As the library grows, some entries may become stale, overly specific, redundant, or misleading for a new task, which can explain non-monotonic trends in zero-shot transfer. More robust retrieval, pruning, ranking, and re-validation mechanisms are needed to keep the library useful at scale. Finally, the debug and evolutionary-search loop is compute-intensive, consuming many LLM calls and simulator or robot rollouts per task; scaling to very large task suites will require either cheaper LLM inference, more sample-efficient search, or stronger mechanisms for reusing prior repairs.

6. Conclusion

We present ASPIRE, a continual learning robotic system that autonomously writes and refines robot control programs while compounding experience into a reusable skill library. ASPIRE operates in an open-ended learning loop with three components: a closed-loop robot execution engine that exposes fine-grained multimodal traces, a continually expanding skill library that distills validated fixes into transferable knowledge, and an evolutionary search procedure that explores diverse task sequences and control programs. Across diverse benchmarks, ASPIRE substantially outperforms existing VLA and coding-agent baselines, demonstrates strong zero-shot transfer to unseen long-horizon tasks, and provides initial evidence that the skills discovered in sim can transfer across embodiment to significantly reduce real-robot programming token cost despite different robot embodiments and APIs.

Acknowledgments

We thank Nadun Ranawaka, Jimmy Wu, Matin Furutan, Haotian Lin, Abhi Maddukuri, Yulu Gan, Matin Nikoui, and Yuqi Xie for their help with open-source support, real robot infrastructure, advice and guidance on the paper, website release, and experimental equipment. We are also grateful to the members of NVIDIA GEAR, UMich SymbioticLab, UsesysLab, UC Berkeley AUTOLab, CMU LeCAR Lab for their kind support.

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A. Skill Library Details

This appendix expands the representative entries shown in Figure 3 into the skill-library taxonomy. Each entry will list: (i) the *problem* extracted from the triggering failure trace; (ii) the *when-to-apply* guard encoding the situational retrieval condition; (iii) the validated *repair snippet*; and (iv) the origin task(s) that produced the entry.

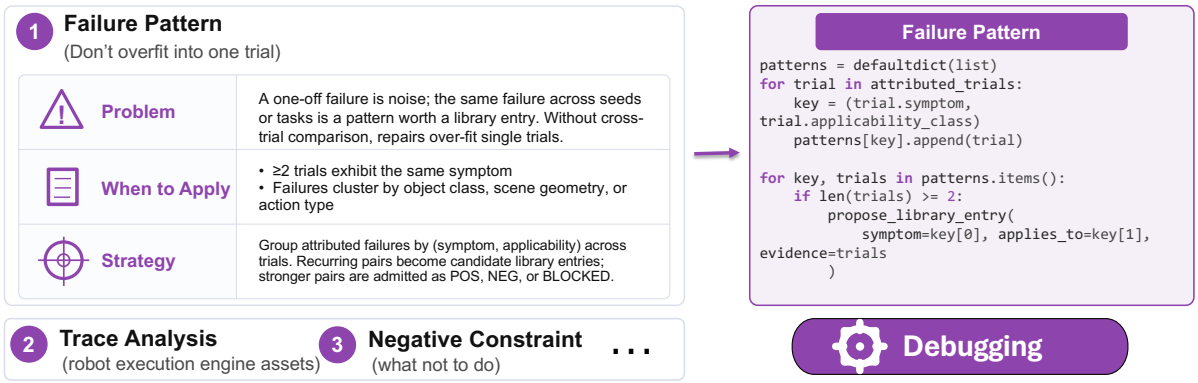


Figure 7: **Debugging skills.** Representative skill-library entries that encode reusable debugging strategies, including failure signatures, when-to-apply guards, and validated repair sketches.

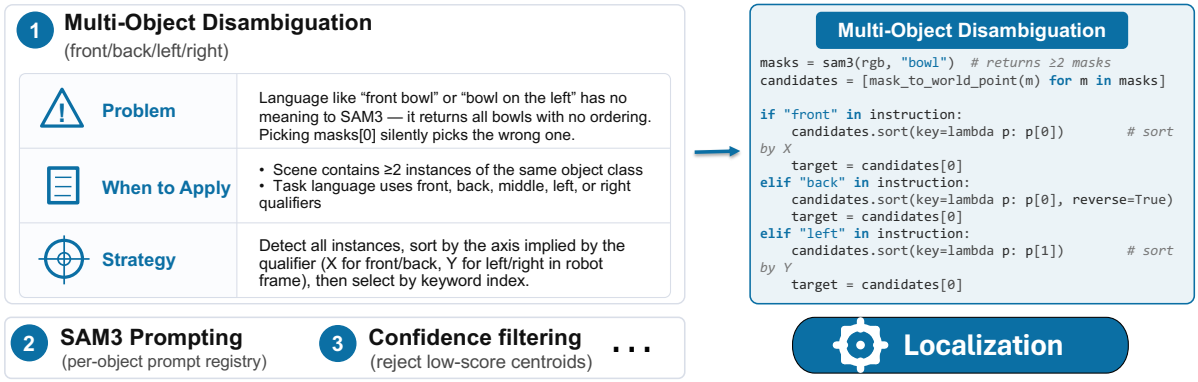


Figure 8: **Localization skills.** Representative entries for grounding ambiguous language and object references into robust perception and localization routines.

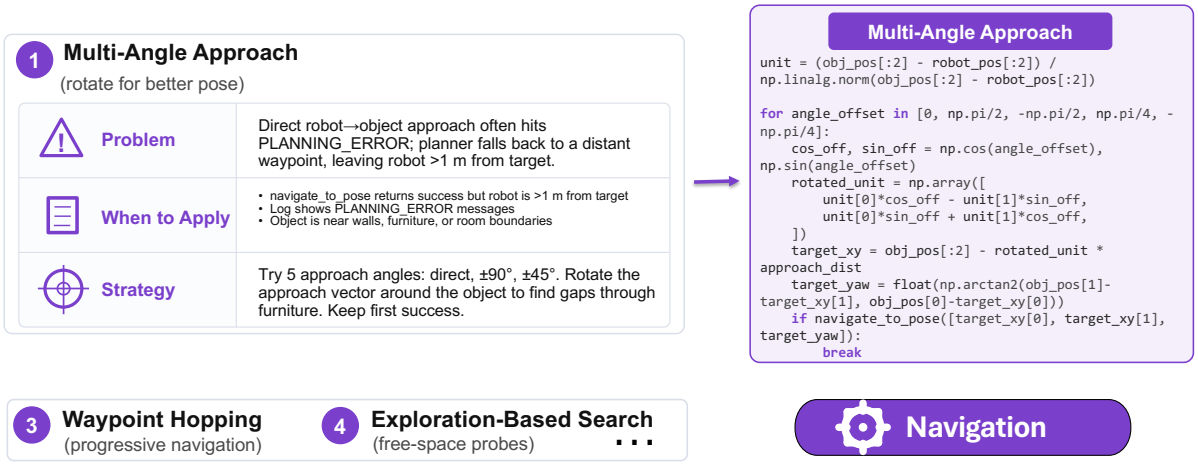


Figure 9: **Navigation skills.** Representative entries for recovering from motion-planning failures and selecting collision-aware approach poses.

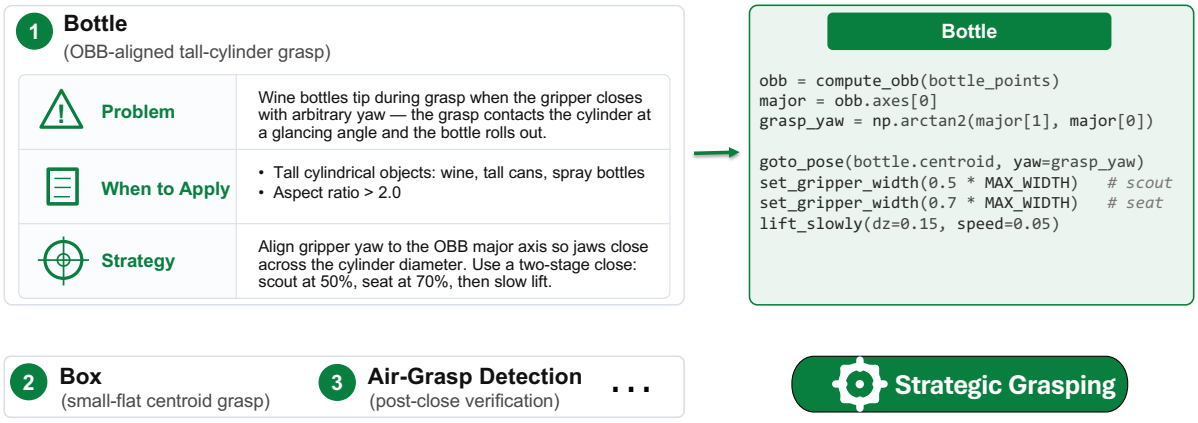


Figure 10: **Strategic grasping skills.** Representative entries for choosing task-appropriate grasp points and adapting grasp strategy to object geometry and scene context.

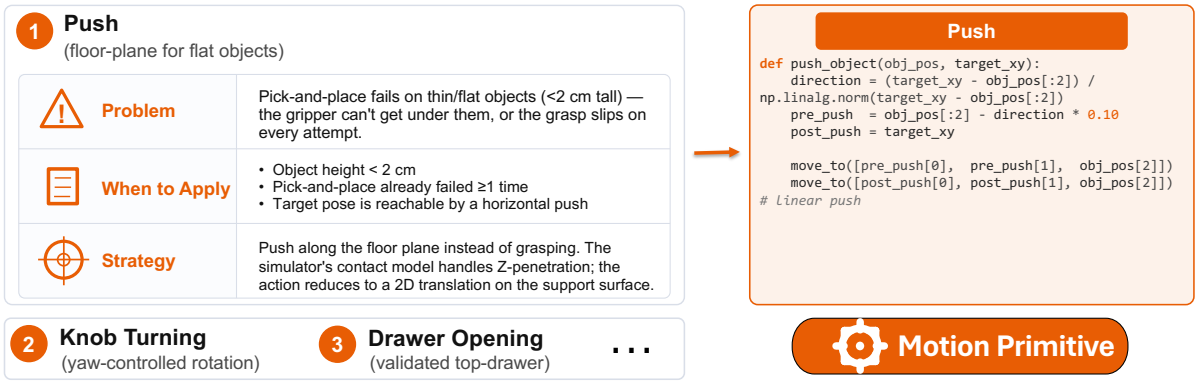


Figure 11: **Motion-primitive skills.** Representative entries for reusable low-level motion patterns, contact-rich alignment, and execution-time recovery.

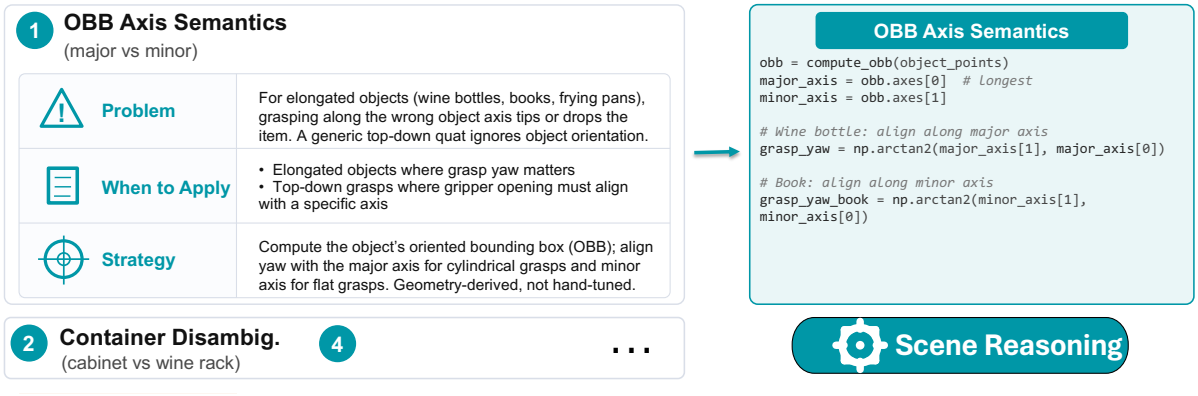


Figure 12: **Scene-reasoning skills.** Representative entries for reasoning over spatial relations, support surfaces, occlusions, and scene-level task constraints.

B. Main Benchmark Result Tables

B.1. Macro-Averaged Main Results

This section reports the exact numerical values visualized in Figure 4 of the main paper.

Table 2: LIBERO-Pro [Zhou et al., 2025] performance of OpenVLA [Kim et al., 2024], π_0 [Black et al., 2025], $\pi_{0.5}$ [Physical Intelligence et al., 2025], CaP-Agent0 [Fu et al., 2026], and ASPIRE on the **libero-object**, **libero-goal**, and **libero-spatial** benchmarks under initial position perturbations (Pos) and instruction perturbations (Task), macro-averaged across 10 tasks per suite. **Overall** columns are macro-averaged across the three suites. All ASPIRE numbers are on the 50-seed held-out evaluation (seeds 1–50). See the per-task breakdown in Appendix D.1.

Method	**libero-object**		**libero-goal**		**libero-spatial**		**Overall**		
	Pos	Task	Pos	Task	Pos	Task	Pos	Task	All
OpenVLA	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
π_0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
$\pi_{0.5}$	0.17	0.01	0.38	0.00	0.20	0.01	0.25	0.01	0.13
CaP-Agent0	0.22	0.18	0.26	0.17	0.12	0.14	0.20	0.16	0.18
Aspire	**0.98**	**0.95**	**0.81**	**0.45**	**0.51**	**0.60**	**0.77**	**0.67**	**0.72**

Table 3: Robosuite [Zhu et al., 2020] performance of CaP-Agent0 [Fu et al., 2026] and ASPIRE on 7 manipulation tasks (100-trial held-out evaluation). Values are task success rates in $[0, 1]$. **Average** is the unweighted mean across all 7 tasks.

Task	**CaP-Agent0**	**Aspire**
cube_lift	**0.97**	**0.97**
cube_stack	0.98	**0.99**
cube_restack	0.89	**1.00**
spill_wipe	**1.00**	0.99
two_arm_handover	0.20	**0.92**
two_arm_lift	**0.74**	0.71
nut_assembly	0.00	**0.09**
Mean	*0.68*	*0.81*

Table 4: BEHAVIOR-1K [Li et al., 2024] performance on two representative household tasks: Soda Can pick-up and Radio pick-up. Navigation and Task Success rates in $[0, 1]$ are reported separately; all numbers are on the 25-seed held-out evaluation (seeds 1–25); ASPIRE runs interactive block-by-block generation with the skill library accumulated on seeds 26–35.

Task	**Human**		**CaP-Agent0**		**Aspire**	
	Nav	Task	Nav	Task	Nav	Task
Soda Can pick-up	0.80	0.72	0.84	0.72	**0.92**	**0.88**
Radio pick-up	0.88	0.36	0.80	0.56	**1.00**	**0.88**

C. LIBERO-Pro Long Zero-Shot Transfer

C.1. Macro-Averaged Main Results

Table 5: Macro-averaged zero-shot transfer of the ASPIRE skill library (accumulated on LIBERO-90) to LIBERO-Pro Long [Zhou et al., 2025]. N denotes the number of LIBERO-90 tasks whose repair skills seed the library; no additional debugging is performed at test time. Columns report macro-averaged success rate on Pos (position perturbation) and Task (instruction perturbation) axes, 10 tasks per axis.

Method	**LIBERO-Pro Long Zero-Shot Transfer**		
	Pos	**Task**	**Overall**
OpenVLA	0.00	0.00	0.00
π_0	0.00	0.00	0.00
$\pi_{0.5}$	0.08	0.01	0.05
CaP-Agent0	0.052	0.024	0.038
ASPIRE (N=0, zero-shot)	0.00	0.094	0.047
ASPIRE (N=25, zero-shot)	0.056	0.218	0.137
ASPIRE (N=50, zero-shot)	0.138	0.292	0.215
ASPIRE (N=90, zero-shot)	**0.226**	**0.383**	**0.305**

C.2. Per-Task Breakdown

Table 6: Zero-shot transfer of the ASPIRE skill library (accumulated on LIBERO-90) to LIBERO-Pro Long, per task and snapshot size N (seeds 1–50, n=50). **Pos**: positional-perturbation variant. **Task**: semantic-perturbation variant.

	Pos				**Task**			
Task	N=0	N=25	N=50	N=90	N=0	N=25	N=50	N=90
Stove + moka pot	0.00	0.30	0.84	**1.00**	0.00	0.02	0.68	0.26
Bowl in drawer	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mug in microwave	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Both mokas on stove	0.00	0.00	0.00	0.00	0.00	0.00	0.84	**1.00**
Soup + cream cheese	0.00	0.12	0.02	0.00	0.00	0.68	0.60	0.70
Soup + tomato sauce	0.00	0.14	0.20	0.00	0.78	0.68	0.02	0.70
Cream cheese + butter	0.00	0.00	0.06	**0.94**	0.00	0.00	0.34	0.86
Mug on two plates	0.00	0.00	0.16	0.32	0.16	0.28	0.08	0.00
Mug + pudding	0.00	0.00	0.06	0.00	0.00	0.00	0.08	0.04
Book in caddy	0.00	0.00	0.04	0.00	0.00	0.52	0.28	0.26
Mean	*0.00*	*0.06*	*0.14*	*0.23*	*0.09*	*0.22*	*0.29*	*0.38*

D. Ablation Details

D.1. LIBERO-Pro Position Perturbation (Swap) Per-Task Breakdown

Table 7: LIBERO-Pro per-task ablation, **Position perturbation** (seeds 1–50, n=50). **Aspire w/o Robot execution engine and Evolutionary search**: zero-shot Claude Opus 4.6 with 15 example programs. **Aspire w/o Evolutionary search**: Robot Execution Engine repaired program (Execution engine + skill library). **Evo. search**: raw evolutionary-search result (best validated candidate), or "–" if Evolutionary Search was not run for that task. **Aspire**: per-task winner selected on a held-out validation set (seeds 66–80) between the repaired program and Evolutionary Search.

Task	w/o Engine & Evo	w/o Evo	Evo. search	Aspire
libero-object				
Alphabet soup	0.00	**1.00**	–	**1.00**
BBQ sauce	0.06	**1.00**	–	**1.00**
Butter	0.00	**1.00**	–	**1.00**
Chocolate pudding	0.00	**1.00**	–	**1.00**
Cream cheese	0.92	**1.00**	–	**1.00**
Ketchup	0.02	**0.92**	–	**0.92**
Milk	0.06	**0.92**	–	**0.92**
Orange juice	0.46	**1.00**	–	**1.00**
Salad dressing	0.18	**0.96**	–	**0.96**
Tomato sauce	0.66	**0.98**	–	**0.98**
Mean	*0.24*	*0.98*	–	*0.98*
libero-goal				
Open middle drawer	0.28	0.02	0.72	**0.72**
Open top drawer+bowl	0.00	**0.86**	–	**0.86**
Push plate to stove	0.00	0.00	0.00	0.00
Bowl on plate	0.04	0.36	0.92	**0.92**
Bowl on stove	0.16	0.56	0.96	**0.96**
Bowl on cabinet top	0.72	0.74	1.00	**1.00**
Cream cheese in bowl	0.00	**0.74**	0.92	**0.74**
Wine on rack	0.06	0.46	0.98	**0.98**
Wine on cabinet top	**1.00**	0.24	1.00	**1.00**
Turn on stove	0.40	0.68	0.90	**0.90**
Mean	*0.27*	*0.47*	*0.82*$^{(n=9)}$	*0.81*
libero-spatial				
Betw. plate+ramekin	0.02	0.10	0.94	**0.94**
From table center	0.00	**0.32**	0.82	**0.32**
In top drawer	0.02	**0.22**	0.10	0.10
Next to cookie box	0.02	**0.16**	0.12	**0.16**
Next to plate	0.08	**1.00**	1.00	**1.00**
Next to ramekin	0.00	**0.20**	0.40	**0.20**
On cookie box	0.00	**0.32**	0.24	0.24
On ramekin	0.58	0.42	0.94	**0.94**
On stove	0.04	**0.86**	–	**0.86**
On wooden cabinet	0.10	**0.56**	0.34	0.34
Mean	*0.09*	*0.42*	*0.54*$^{(n=9)}$	*0.51*
Overall mean	*0.20*	*0.62*	*0.68*$^{(n=18)}$	*0.77*

D.2. LIBERO-Pro Task Perturbation (Goal) Per-Task Breakdown

Table 8: LIBERO-Pro per-task ablation, **Task perturbation** (seeds 1–50, n=50). **Aspire w/o Robot execution engine and Evolutionary search**: zero-shot Claude Opus 4.6 with 15 example programs. **Aspire w/o Evolutionary search**: Robot Execution Engine repaired program (Execution engine + skill library). **Evo. search**: raw evolutionary-search result (best validated candidate), or "–" if Evolutionary Search was not run for that task. **Aspire**: per-task winner selected on a held-out validation set (seeds 66–80) between the repaired program and Evolutionary Search.

Task	w/o Engine & Evo	w/o Evo	Evo. search	Aspire
libero-object				
Alphabet soup	0.00	**1.00**	–	**1.00**
BBQ sauce	0.88	**1.00**	–	**1.00**
Butter	0.78	**0.98**	–	**0.98**
Chocolate pudding	0.00	**1.00**	–	**1.00**
Cream cheese	0.00	**0.72**	0.74	**0.72**
Ketchup	0.00	**0.96**	–	**0.96**
Milk	0.00	**1.00**	–	**1.00**
Orange juice	0.00	**1.00**	–	**1.00**
Salad dressing	0.00	**0.92**	–	**0.92**
Tomato sauce	0.04	**0.96**	–	**0.96**
Mean	*0.17*	*0.95*	–	*0.95*
libero-goal				
Open middle drawer	0.00	0.00	0.00	0.00
Open top drawer+bowl	0.00	**0.50**	0.46	0.46
Push plate to stove	0.26	0.58	0.76	**0.76**
Bowl on plate	0.00	**0.88**	0.82	**0.88**
Bowl on stove	0.00	0.00	0.00	0.00
Bowl on cabinet top	0.00	0.00	0.04	**0.04**
Cream cheese in bowl	0.00	**0.82**	0.74	0.74
Wine on rack	0.00	0.74	0.86	**0.86**
Wine on cabinet top	0.00	**0.14**	0.72	**0.14**
Turn on stove	0.24	0.52	0.66	**0.66**
Mean	*0.05*	*0.42*	*0.51*$^{(n=10)}$	*0.45*
libero-spatial				
Betw. plate+ramekin	**0.14**	0.08	0.84	0.08
From table center	0.26	0.46	0.64	**0.64**
In top drawer	0.04	**0.96**	–	**0.96**
Next to cookie box	0.00	**0.94**	–	**0.94**
Next to plate	0.08	**0.70**	0.92	**0.70**
Next to ramekin	0.00	**0.60**	0.42	**0.60**
On cookie box	0.00	0.06	0.70	**0.70**
On ramekin	0.20	0.10	0.46	**0.46**
On stove	0.00	0.32	0.42	**0.42**
On wooden cabinet	0.13	**0.46**	0.98	**0.46**
Mean	*0.04*	*0.47*	*0.67*$^{(n=8)}$	*0.60*
Overall mean	*0.09*	*0.61*	*0.59*$^{(n=19)}$	*0.67*

D.3. Evolutionary Search Progress Per-Task Breakdown

Table 9: Evolutionary search progress on selected LIBERO-Pro tasks. Each column shows the held-out success rate (seeds 1–50) of the best candidate at that iteration. Blank cells indicate search terminated early.

Suite	**Task**	**Iter 0**	**Iter 1**	**Iter 2**	**Iter 3**	**Iter 4**
goal_swap	Bowl → plate	0.62	0.60	0.60	0.18	**0.86**
goal_swap	Wine bottle → rack	0.40	**0.76**	0.74		
goal_swap	Bowl → stove	0.62	**0.82**			
goal_task	Push plate → stove	0.00	**0.80**			
spatial_swap	Bowl on cabinet → plate	0.16	0.38	0.38	**0.46**	0.30
spatial_swap	Bowl next to cookie → plate	0.16	0.20	0.04	0.04	**0.40**
spatial_task	Bowl on cookie → plate	0.26	0.04	**0.70**		
spatial_task	Bowl on ramekin → plate	0.02	0.00	0.22	**0.36**	0.18

E. Agent Pipeline Skills

E.1. Skill Library Prompts

Actor agents report candidate repairs using a structured findings schema: the observed failure mode, the validated repair, transferable patterns, task-specific quirks, and the validation success rate on debug configurations. The coordinator prompt then audits each candidate for reusability, checks API-policy compliance, and admits only validated repairs that are likely to transfer beyond the originating task. Parallel actor agents are instructed to write task-level repairs and findings, while the coordinator serializes skill admission to avoid conflicting library writes. See subagent skills in E.3 and E.4 for exact formats. See code release for all agent skill files.

E.2. Agent System Prompt and Persistent Memory

CLAUDE.md

```
# CLAUDE.md
This file is the constitution for Claude Code working in this repository. Read it fully before doing
    anything.

## STARTUP: Read Project Memory First

At the start of every session, **immediately read** `.claude/memory/MEMORY.md` before doing anything else.
     It contains project vision, key conventions, and the skills index.

## What Is ASPIRE

ASPIRE is a framework where LLMs write Python code to control robots through a curated tool API. Code is
    executed in a sandboxed simulator, the simulator verifies success, and the agent improves through
    failure diagnosis and skill accumulation.

## FORBIDDEN APIs

**These APIs access simulator ground truth and are STRICTLY FORBIDDEN in all fix code, debug scripts, and
    skill implementations.** Using them invalidates benchmark results, as they don't transfer to real
    robots.

```
[FORBIDDEN] env.handle.env.sim - no MuJoCo sim object access
[FORBIDDEN] sim.data.body_xpos - no ground-truth object positions
    ... (rest of APIs omitted)
```
**Rule of thumb:** If a real robot with a camera could do it, it's allowed. If it reads the physics engine
    's internal state, it's forbidden.

**Also forbidden:** Reading simulator asset files (`.bddl`, `.xml`, `.urdf`, MuJoCo model files) to infer
    object geometry, success predicates, or scene structure. Treat these as inaccessible - diagnose
    purely from observations and traces.

## ALLOWED APIs

Full source: `integrations/franka/libero_reduced_skill_library.py` (and `libero_reduced.py` for base class
    ). Read the source for exact signatures, return types, and edge cases.
```
[ALLOWED] get_observation()   - RGB, depth, intrinsics, extrinsics, robot state
[ALLOWED] segment_sam3_text_prompt(rgb, text)   - SAM3 text-prompted segmentation
    ... (rest of APIs omitted)
```
**You are encouraged to design new helper functions** (e.g. `localize_object()`, `make_topdown_quat()`)
    when you find yourself repeating patterns. Add them to the relevant skill in `.claude/skills/` so
    future sessions can reuse them.

## Skills
```

```
All skills live in `.claude/skills/<name>/SKILL.md` and are auto-discovered. Read the relevant skill
    before starting work.

| Skill | What it covers |
|---|---|
    ... (skills list omitted)
| **add more as needed** | **document new patterns here as you discover them** |

## Log Everything

Append to `./docs/logs/YYYY-MM-DD.md` after any significant work: experiments run, fixes tried, skills
    updated, results observed. Be detailed - include trial numbers, success rates, error messages, file
    paths.
```

MEMORY.md

```
# ASPIRE Project Memory

## What We're Building
A Self-Improving Coding Agent for robot manipulation. The agent debugs its own failures, accumulates
    reusable skills, and improves its perception tools.

- Foundation: Rich debugging playground (traces, REPL)
- Layer 1: Auto-growing skill library
- Layer 2: Program search (Pass@K parallel search)

## Key Conventions
- **NEVER git commit unless user explicitly asks**, do NOT auto-commit after skill updates or log writes
- **NEVER read sim asset files** (`.bddl`, `.xml`, `.urdf`, MuJoCo models), diagnose from observations and
     traces only
- **Always write reusable analysis as scripts** (not inline `python3 -c "..."`) and reference them in the
    relevant skill
    ... (rest of conventions omitted)

## Skills Available
    ... (skills list omitted)
```

api-reference.md

```
## Observation and Perception

| Function | Returns |
|---|---|
| `get_observation()` | RGB image, depth image, camera intrinsics, camera pose, and robot state. |
| `segment_text_prompt(rgb, text_prompt)` | A list of masks with bounding boxes, scores, and labels. |
| `segment_point_prompt(rgb, point_coords)` | A list of masks from point-conditioned segmentation. |
| `point_prompt_vlm(image, text_prompt)` | A pixel coordinate for the prompted object or region. |

## Grasp Planning and IK

| Function | Returns |
|---|---|
| `plan_grasp(depth, intrinsics, segmentation)` | Candidate grasp poses and scores in the camera frame. |
| `select_top_down_grasp(poses, scores, ...)` | The best top-down grasp transform and score. |
| `solve_ik(position, quaternion_wxyz)` | A 7-DoF joint configuration, or failure if unreachable. |

Camera-to-world conversion is required after grasp planning.

```python
camera_to_world = obs["agentview"]["pose_mat"]
grasp_world = camera_to_world @ grasp_camera
```
```

````
## Motion Execution

| Function | Returns |
|---|---|
| `move_to_joints(joints)` | Blocking joint-space motion. |
| `open_gripper()` / `close_gripper()` | Gripper command. |
| `goto_home_joint_position()` | Move to a predefined safe home configuration. |

## Geometry Utilities

| Function | Description |
|---|---|
| `decompose_transform(T)` | Convert a 4x4 transform into position and quaternion. |
| `mask_to_world_points(mask, depth, intrinsics, extrinsics)` | Project a 2D mask into a world-frame point
      cloud. |
    ... (rest omitted)

## Task Language

```python
env.task_language
```

The runtime task language is the reliable source of the actual goal, especially when benchmark filenames
     do not match remapped task instructions.

## Success Conditions

| Placement Type | Tolerance |
|---|---|
| Regular object placement | Tight XY tolerance plus contact. |
| Site-object placement | Larger site-local tolerance. |

## Scene Coordinate Frames

Use observation-derived depth projection for object positions and avoid hard-coded scene coordinates.
Reference frame constants are environment-specific and should not be used as object localization shortcuts.


`... (benchmark-specific table of scene constants omitted) ...`

## Output Directory Structure

```text
<OUTPUT_DIR>/
  <suite>/<task>/<run_id>/
    run/
      <done_flag>
      trial_<seed>_<attempt>_reward_<score>_taskcompleted_<flag>/
        code.py
        trace.json
        keyframes/
        video_combined.mp4
        visual_feedback.png
        summary.txt
        all_responses.json
```

The trial folder name records the seed, attempt index, reward, and task-completion flag.
The attempt index distinguishes clean exits from crashes, and `reward` is the scalar task score.

## Trace Format

```json
{
  "step": 5,
  "function": "segment_text_prompt",
````
````

```
  "args": {"rgb_shape": [512, 800, 3], "text_prompt": "bowl"},
  "duration_ms": 1463.8,
  "result": {
    "num_masks": 200,
    "mask_0_score": 0.917,
    "mask_0_bbox": [369.2, 257.1, 438.4, 310.5],
    "mask_0_area_pct": 0.68
  },
  "keyframe_saved": true
}
```

Key Source Files

File	Role
`<batch_runner>`	Runs suites, tasks, and model configurations.
`<trial_loop>`	Resets the environment, executes generated code, and saves artifacts.
`<single_trial_runner>`	Replays saved code or runs a fresh trial.
`<single_trial_analyzer>`	Performs structured trace analysis for one trial.
`<multi_candidate_trace_analyzer>`	Performs per-candidate trace analysis for evolutionary search runs.
`<progress_scanner>`	Aggregates success rates across an output directory.
`<robot_api_source>`	Defines the robot control and geometry API.
`<trace_logger_source>`	Records API calls, return values, timings, and keyframes.
`<evaluation_config>`	Defines benchmark and replay configuration.

E.3. ASPIRE Fix Loop Skills

coordinator.md

```
> **Purpose:** Coordinate repair of failed robot trials from a baseline policy and validate finalized
    fixes on a held-out seed set.
> **Baseline results:** `&lt;BASELINE_OUTPUT_DIR&gt;/`
> **Progress tracker:** `&lt;PROGRESS_FILE&gt;`
> **Subagent prompt:** `&lt;SUBAGENT_PROMPT_TEMPLATE&gt;`

### Initialization: Verify Perception Services

Before dispatching subagents, confirm that all required perception services are running. A service-
    specific health check should return the expected "up" status.

If any service is down, start it before proceeding. Subagents may fail silently without these dependencies.


### Main Loop

```text
read progress → assign free devices → dispatch subagents → go idle
                                                           ^
on notification: update progress/skills + redispatch device --|
```

One task is handled by one subagent on one compute device. Repeat until no eligible tasks remain.

Coordinator Rules

1. **Dispatch subagents rather than debugging directly.** The coordinator assigns work, waits for
    completion notifications, updates shared knowledge, and dispatches the next task.
2. **Go idle after dispatching.** Completion notifications indicate when a device is free; avoid polling
    or intervening while subagents are running.
3. **Keep available devices occupied.** Any free device should receive the next pending task.
4. **Do not inspect task-specific debugging artifacts.** Files such as traces, generated programs,
    summaries, or keyframes are the subagent's responsibility. The coordinator may read the subagent's
    final findings when updating shared skills.
```

````
5. **Never re-dispatch completed tasks.** Completed validation tasks must not be run again, because
     duplicate trial directories can corrupt aggregate results.

### Workflow

#### 1. Read the Progress Tracker

```bash
cat <PROGRESS_FILE>
```

Identify tasks marked `pending` or `stage1-complete`.

#### 2. Check Free Compute Devices

Assign tasks only to devices that are free. Never assign two tasks to the same device.

#### 3. Dispatch One Subagent per Free Device

```python
Agent(
    description="Fix loop: <suite>/<task>",
    subagent_type="general-purpose",
    prompt=<filled_subagent_prompt>,
    run_in_background=True,
)
```

Send all dispatches together, then stop. The coordinator resumes when completion notifications arrive.

#### 4. On Completion: Redispatch and Update Skills

When a subagent completes, its result should include the device identifier. The coordinator then:

1. Reads the progress tracker.
2. Dispatches the next eligible task on the freed device.
3. Reads the completed task's `findings.md`.
4. Promotes generalizable patterns into the shared skill library.
5. Goes idle again.

If the result does not include a device identifier, query the device monitor to infer which device is free.


#### 5. Update Shared Skills

After each completion, read:

```text
<BASELINE_OUTPUT_DIR>/<SUITE>/<TASK>/findings.md
```

Promote only generalizable patterns to the skill library.

`... (skill-library category table omitted) ...`

### Validation Runs

For pure validation runs where final repair code already exists, direct scripts are more reliable than
     autonomous subagents. The validation script should run the held-out seed set exactly once and write
     to the official output directory.

`... (task-specific validation script, log parsing, and environment variables omitted) ...`

Important rule: pass only the validation base directory if the replay script appends `<suite>/<task>`
     internally.

### Suite Name Collision Warning
````

````
Some benchmark suites may contain identical task names under different perturbation conditions. Always
     include the full suite name when briefing a subagent.

### Perturbation Notes

- Position-perturbation suites randomize object poses by seed; perception should handle this naturally.
- Language-perturbation suites remap the task goal; the executable task
  instruction should be read from the runtime environment rather than inferred from a filename.
````

subagent.md

````
The following prompt is filled with one suite, task, and device assignment before being passed to a
     background subagent.

```text
## Task Assignment

SUITE: <SUITE_NAME>
TASK:  <TASK_NAME>
DEVICE: <DEVICE_ID>

Working directory: <PROJECT_ROOT>
```

### Evaluation-Set Lockout

The held-out validation seeds are locked during debugging.

- During Stage 1, replay only the debugging seeds.
- Do not replay validation seeds for testing, tuning, temporary experiments, or alternate code versions.
- Stage 2 uses the validation seeds exactly once, after `fix_code.py` is finalized, and writes to the
     official validation output directory.
- Stage 2 must run to completion before the subagent returns.

Violating this rule invalidates the benchmark result.

### Role

You are a task-level repair subagent for a robotics benchmark. Your job is to diagnose failed robot trials
      for one task, write a generalizable repair program, validate it according to the protocol, and
     report concise results.

### Context

The system asks language models to write Python control programs for a robot arm through a curated
     perception and manipulation API. Programs run in simulation, and failures are diagnosed through
     traces, summaries, and visual keyframes. The baseline policy has already run a set of debugging seeds
     ; your job is to repair the task program without using simulator ground truth.

#### Forbidden APIs

The repair program must not access simulator internals, ground-truth object states, hidden task predicates
     , asset files, or low-level physics state. Examples include:

```text
simulator body positions
site positions
`... (additional forbidden APIs omitted) ...`
```

These APIs would not transfer to a real robot and invalidate the result.

#### Allowed APIs

The repair code may use only observation-derived perception, manipulation primitives, and standard
     numerical computation:
````

```
```text
get_observation()
segment_text_prompt(rgb, text)
`... (additional allowed APIs omitted) ...`
```

Always use the project virtual environment or pinned runtime, not the system Python.

Before any replay, verify that the required perception services are running.

Stage 1: Debug on Non-Held-Out Seeds

Step 0: Check Whether Repair Code Already Exists

If it exists, skip Stage 1 and check whether Stage 2 has already completed.

Step 1: Fast Path from Successful Baseline Trials

First, check whether any baseline debugging seed already succeeded:

If successful trials exist:

1. Read the successful generated program.
2. Extract the final code block if multiple attempts are present.
3. Test it on a small number of failed debugging seeds.
4. If it succeeds, save it as the task-level `fix_code.py` and proceed to Stage 2.
5. If it fails, continue to the full debug loop.

If no debugging seeds succeeded, continue directly to the full debug loop.

`... (exact replay command and temporary output paths omitted) ...`

Step 2: Diagnose Failed Debugging Seeds

For each failed seed, inspect the last attempt for that seed:

- `trace.json`: function calls, return values, perception failures, IK failures,
  and gripper signals.
- `code.py`: the baseline strategy.
- `keyframes/`: visual observations at perception steps.
- `summary.txt`: stdout, stderr, and reward.

Interpretation examples:

```text
large gripper width after closing → likely object grasped
medium gripper width → marginal grasp
small gripper width → likely air grasp
zero masks → poor perception prompt or occlusion
missing IK solution → target outside reachable workspace
early crash → inspect stdout/stderr summary
```

Step 3: Write and Test Repair Code

Before writing code, read the relevant skill-library entries for localization, grasping, transport, and
    manipulation patterns.

Write `fix.py` from the diagnosis and replay it on debugging seeds only.

If the repair fails, read the new trace and iterate. A live inspection REPL may be used for observation-
    derived checks such as segmentation counts.

`... (interactive REPL command and shell-specific buffering notes omitted) ...`

Hard limit: at most three replay attempts per debugging seed. Any run against a seed counts as an attempt,
     regardless of output directory or code version. After the cap, write a `BLOCKED.md` report and move
```

```
    on.

‘BLOCKED.md‘ format:

‘... (blocked-seed report template omitted) ...‘

#### Step 4: Synthesize the Task-Level Repair Program

After debugging all seeds, synthesize one generalizable ‘fix_code.py‘. The code must not contain seed-
    specific branches.

Save it to ‘<BASELINE_OUTPUT_DIR>/<SUITE>/<TASK>/fix_code.py‘. Also write <BASELINE_OUTPUT_DIR>/<SUITE>/<
    TASK>/findings.md‘.

Use this findings format:

‘‘‘markdown
## Task: <SUITE> / <TASK>

### Root Cause(s)
- <concise description of each failure mode>

### What Fixed It
- <change that resolved each root cause>

### Perception Prompts That Worked
| Object | Prompts (priority order) | Notes |
|---|---|---|
| <object> | "<prompt1>", "<prompt2>" | <caveats> |

### Generalizable Patterns
- <patterns likely to apply beyond this task>

### Task-Specific Quirks
- <details specific to this task>

### Stage 2 Success Rate
<N>/<TOTAL_VALIDATION_SEEDS>
‘‘‘

If all debugging seeds are blocked, skip Stage 2 for this task.

### Stage 2: Validate on Held-Out Seeds

Stage 2 is a one-shot validation operation. Do not debug Stage 2 runs.

Before running, check whether validation has already completed:

If enough unique validation seeds are already present, do not re-run Stage 2. Report the existing results
    and return. Otherwise, run validation sequentially:

‘... (validation loop, log parsing, success counting, and environment variables omitted) ...‘

Verify that outputs landed in the expected directory before reporting results.

### Final Action

Run the shared progress-generation script as the last action. It should update the progress tracker
    atomically and be safe under concurrent subagent execution.

### Return Format

Return only a brief structured summary:

‘‘‘text
SUITE: <SUITE>
TASK: <TASK>
DEVICE: <DEVICE>
```

```
Stage 1: fix_code.py written: yes/no
  Seeds fixed: <count or list>
  Seeds blocked: <count and one-line cause>

Stage 2: <N>/<TOTAL_VALIDATION_SEEDS>
  Output dir: <VALIDATION_OUTPUT_BASE>/<SUITE>/<TASK>/

Key findings:
  - <generalizable root cause and fix>
  - <useful perception prompt, if notable>
  - <pattern that should inform future tasks>
```

E.4. ASPIRE Evolutionary Search Skills

Evolutionary search (Section 2.3) maintains a persistent task-analysis document A_i that is carried across rounds. A_i consists of (i) a per-task scene description populated once from an initial snapshot (object shapes, goal geometry, obstacles, blocked approach directions), (ii) running hypotheses and the candidate metadata that tests them, and (iii) a ledger of *eliminated* directions (through testing) and *blocked, untested* directions (failed due to a workspace constraint). UpdateAnalysis rewrites A_i each round from the new traces, keyframes, and library retrievals. This document is what lets a later round avoid re-testing an eliminated branch while still retrying blocked branches once a new technique (e.g. a wrist rotation) becomes available.

coordinator.md

```
> **What:** Run Evolutionary search iterative debugging with multiple candidates per iteration on a list
    of low-performing tasks.
> **Why:** Intensive debugging for difficult tasks; multi-candidate search can find strategies the single-
    pass actor misses.
> **Subagent template:** `<EVOSEARCH_SUBAGENT_PROMPT_TEMPLATE>`

### Initialization: Verify Perception Services

Before dispatching subagents, verify that all required perception services are running.
All services must be up before dispatching.

`... (service health-check command and machine-specific ports omitted) ...`

### Directory Layout

```text
<EVOSEARCH_OUTPUT_DIR>/
  <suite>/<task>/<run_id>/
    task_analysis.md
    iter_00/ iter_01/ ...
      candidate_A/code.py
      iter_summary.json
  <suite>/<task>/
    evosearch_best_code.py
    findings.md

<VALIDATION_OUTPUT_DIR>/
  <suite>/<task>/... (rest omitted)
```

Task List

Run `<progress_generation_script>` to verify exact task names before dispatch.
```

````
Tasks below the chosen baseline success threshold are candidates for repair.

`... (experiment-specific task ordering and threshold table omitted) ...`

### The Loop

```text
read progress → assign free devices → dispatch subagents → go idle
                                                            ^
on notification: redispatch freed device to next pending task -|
```

One task = one subagent = one compute device.
The subagent runs iterations on the debug set and then runs Stage 2 on the held-out set before returning.
When the coordinator gets a completion notification, the device is already free; dispatch the next task.

### Coordinator Rules

1. **Dispatch subagents -- never run iterations yourself.**
2. **Go idle after dispatching.** You will be notified when a subagent finishes.
3. **Keep available devices occupied.**
4. **On each notification: check which device freed up, dispatch next pending task to it.**
5. **Never re-dispatch a done task.** `done` means Stage 2 complete with enough unique held-out seeds on
   disk.

### Workflow

#### 1. Check Progress

Check which tasks have a saved `evosearch_best_code.py` and how many held-out seeds have already been
   evaluated.

`... (progress-check shell loop omitted) ...`

#### 2. Check Free Compute Devices

Assign tasks only to devices that show no active jobs.

`... (device-monitoring command omitted) ...`

#### 3. Dispatch Subagents

Use the stronger reasoning model selected for the experiment.

```python
Agent(
    description="Evolutionary search actor: <suite_short>/<task_short> device<N>",
    subagent_type="general-purpose",
    model="<HIGH_CAPACITY_MODEL>",
    prompt=<filled_evosearch_subagent_prompt>,
    run_in_background=True,
)
```

Send all dispatches in one message, one per free device, then stop.

#### 4. On Each Completion: Redispatch

When a subagent notification arrives, it has already completed both Stage 1 and Stage 2.
Read `<EVOSEARCH_OUTPUT_DIR>/<SUITE>/<TASK>/findings.md`, note Stage 1 and Stage 2 rates, check free
   devices, dispatch the next pending task, and go idle.

### Stopping Criteria Reference

Subagents stop Stage 1 when the best candidate reaches the success threshold on debug seeds, the maximum
   number of iterations is reached, or the task is blocked.
If Stage 1 is blocked, the subagent skips Stage 2 and returns immediately.
````

subagent.md

````
The following prompt is filled with one suite, task, device assignment, baseline rate, and output location
     before dispatch.

```text
## Task Assignment

SUITE: <SUITE>
TASK: <TASK>
DEVICE: <DEVICE_ID>
TASKSHORT: <SHORT_UNIQUE_NAME_FOR_LOGS>
BASELINE_RATE: <BASELINE_RATE>
EVOSEARCH_DIR: <EVOSEARCH_OUTPUT_DIR>
EVALDIR: <VALIDATION_OUTPUT_DIR>

Working directory: <PROJECT_ROOT>
```

### Evaluation-Set Lockout During Iterations

The held-out evaluation seeds must not be run during iterations.
All iteration evaluations use only the debug seed set.
Stage 2 uses the held-out seeds after iterations converge and the final code is chosen.
Violation invalidates the benchmark.

### Role

You are a debugging subagent.
Your job is to run evolutionary search iterative debugging on debug seeds until convergence or plateau,
     save the best code, run Stage 2 on held-out seeds, and return a structured findings report with both
     Stage 1 and Stage 2 results.
Use the project virtual environment or pinned runtime, never the system Python.

### Context

The actor pipeline achieved `$BASELINE_RATE` on this task.
Your target is to exceed that significantly, ideally reaching the Stage 1 success threshold on debug seeds.

Read the skill library before writing candidates.
Verify that the required perception services are running before replay.

`... (experiment-specific config path and service-port checks omitted) ...`

### Forbidden APIs

```text
simulator body positions
site positions
`... (rest of forbidden APIs omitted)
```

### Stage 1: Evolutionary Search Iterations on Debug Seeds

#### Step 0: Check Whether Best Code Already Exists

If `evosearch_best_code.py` exists, skip to the return step.

#### Step 0b: Task-Language Remapping Check

The task name might not match the actual goal.
Always check and print the runtime task language.
If it differs from `$TASK`, the runtime task language is ground truth; base all strategy on it and record
     the actual language in `task_analysis.md`.

#### Step 1: Read Skills and Existing Baseline
````

```
Read skill-library entries for grasping, localization, transport, manipulation, and any task-specific
     companion skills.
Also read existing repair code if it exists.
If baseline code is found, record its path in ‘task_analysis.md‘ and seed it as ‘candidate_A‘ verbatim.

‘... (skill-file reads and baseline fallback command omitted) ...‘

#### Step 2: Create Run Directory and Scene Snapshot

Create a timestamped run directory under ‘<EVOSEARCH_OUTPUT_DIR>/<SUITE>/<TASK>/‘.
Before writing candidates, run a scene snapshot on one debug seed and inspect the wide and wrist-view
     images.
Populate ‘task_analysis.md‘ with object shape, grasp strategy, goal geometry, placement strategy,
     hypotheses for the first iteration, and an iteration log.

‘... (snapshot command and task_analysis.md template omitted) ...‘

#### Step 3: Write K Candidates for the First Iteration

Always seed ‘candidate_A‘ from the existing baseline repair code if it exists.
Each candidate must test a distinct hypothesis.
No two candidates should fail at the same stage for the same reason.
Each candidate should include a docstring describing the hypothesis, how it differs from prior candidates,
      and the expected failure mode if wrong.

#### Step 4: Evaluate the Iteration

Evaluate all candidates on the fixed debug seed set.
Use the same debug seeds across all iterations so cross-iteration comparison is fair.
Read the leaderboard after completion and analyze traces.

‘... (multi-candidate evaluation command, leaderboard parser, and trace analyzer command omitted) ...‘

#### Step 5: Iterate

Stop Stage 1 when any stopping criterion is met.
When progress stalls, use the remaining iteration budget to explore structurally new approaches rather
     than stopping early.
A plateau on the current approach family means the current family may be wrong, not that the task is
     unsolvable.

The debug seeds are a small, potentially unrepresentative sample.
The code will be evaluated on unseen held-out seeds; write for those, not for the debug seeds.
Prefer strategies that work for mechanistic reasons over strategies that exploit patterns specific to
     debug-seed failures.
Avoid hard-coded thresholds, image-region masks, or offsets derived by fitting to observed debug-seed
     failures.

Per iteration:

1. Update ‘task_analysis.md‘ with new geometry from keyframes.
2. Append an iteration log with the leaderboard, eliminated hypotheses, and open questions.
3. Write K new candidates seeded from the top survivors of the current iteration.
4. Evaluate all candidates on the same debug seed set.
5. Read trace-analysis output and watch for arm blocking, gripper-width signals, perception failures, and
     IK failures.

#### Step 6: Save Best Code, Run Stage 2, Write Findings

Save the best final candidate as ‘<EVOSEARCH_OUTPUT_DIR>/<SUITE>/<TASK>/evosearch_best_code.py‘.
If the best candidate does not beat the baseline candidate on debug seeds, fall back to the stronger
     baseline so Stage 2 uses the best available code.
If all approaches fail, mark the task blocked and skip Stage 2.

‘... (best-candidate sanity check and copy commands omitted) ...‘

### Stage 2: Evaluate Best Code on Held-Out Seeds
```

````
Run the selected best code on each held-out seed exactly once.
Do not use Stage 2 results to revise the code.
Poll until the validation loop completes, then read final success counts.

`... (detached validation script, polling commands, and final count commands omitted) ...`

### Findings Report

Write `<EVOSEARCH_OUTPUT_DIR>/<SUITE>/<TASK>/findings.md`:

```markdown
## Task: <SUITE> / <TASK>
## Baseline rate: <BASELINE_RATE>
## Best evosearch rate on debug seeds: <N>/<DEBUG_TOTAL> (<pct>%)
## Stage 2 on held-out seeds: <N>/<HELD_OUT_TOTAL> (<pct>%)

### Evolutionary Search Run
- Run dir: <RUN_DIR>
- Iterations completed: <N>
- Stopping reason: <solved|max_iterations|blocked>

### What Fixed It vs Baseline
- <key strategy change>

### Perception Prompts That Worked
| Object | Prompts | Notes |
|---|---|---|

### Failure Modes Eliminated
- <approach that failed and why>

### Generalizable Patterns
- <anything worth adding to the skill library>

### Skill Library Updates Made
- <if any skills were updated during this run>
```

### Return Format

```text
SUITE: <suite>
TASK: <task>
DEVICE: <N>
Baseline rate: <from actor pipeline>

Stage 1:
  Best candidate: <candidate_X> at iter_NN
  Best pass rate: <N>/<DEBUG_TOTAL> (<pct>%)
  Stopping reason: <solved|max_iterations|blocked>
  Iterations run: <N>
  Run dir: <RUN_DIR>

Stage 2: <N>/<HELD_OUT_TOTAL> (<pct>%)

Key findings:
  - <strategy that worked vs baseline failure mode>
  - <perception prompts or grasp parameters>
  - <generalizable pattern for the skill library>
```
````

E.5. Initial Skill Library Templates

grasp initial SKILL.md

````
---
name: grasp
description: Structural template for pick-and-place programs - topdown-quat construction, pre-grasp
     approach, lower-close sequence, lift-transport-place pattern. No task-specific parameters; those grow
      through experiment.
---

# Grasp - Structural Template for Pick-and-Place

> **Purpose:** Code skeletons that every pick-and-place task reuses. No task-specific z-offsets or yaws -
     those go in the per-task `task_code.py`.
> **Ownership:** Coordinator adds generalizable patterns. Subagents do not edit this file.

---

## Top-Down Quaternion

```python
import numpy as np
from scipy.spatial.transform import Rotation

def make_topdown_quat(yaw_deg=0):
    """Build a top-down end-effector quaternion (xyzw→wxyz convention used by solve_ik)."""
    R = Rotation.from_euler('z', yaw_deg, degrees=True).as_matrix() @ \
        np.array([[1, 0, 0], [0, -1, 0], [0, 0, -1]])
    q = Rotation.from_matrix(R).as_quat()  # scipy returns xyzw
    return np.array([q[3], q[0], q[1], q[2]])  # reorder to wxyz
```

**Yaw selection:** default 0. If the grasp fails because the gripper fingers collide with a neighboring
     object, try 90 or 45 degrees.

---

## Standard Pick-and-Place Skeleton

```python
# Assume: obj_center, obj_mask, tgt_center, tgt_pts already localized (see localize skill)
# Assume: depth, K (intrinsics), E (camera extrinsic pose_mat) from get_observation()

# 1. Plan grasp via GraspNet on object mask
# 2. Pre-grasp approach: gripper above object, open
# 3. Lower to grasp pose, close gripper
# 4. Lift (avoid dragging object sideways into neighbors)
# 5. Move above target (same z as lift)
# 6. Lower to release height (surface_z + margin)
# 7. Release
# 8. Let physics settle (important for drop-success reading)
    ... (skeleton code omitted)
```

---

## Why `select_top_down_grasp`?

`plan_grasp` returns grasps in camera frame, often angled or side-approach. We want top-down for table-top
      pick-and-place. The selector transforms to world frame and filters for top-down direction (z-axis
     roughly anti-aligned with gravity). When it returns None (no valid top-down), fall back to `E @
     grasp_poses[grasp_scores.argmax()]`.

---

## When to NOT Use This Template
````

```
- **Shelves / drawers / microwaves** - use `manipulation` skill, not this.
- **Stacking** - need stacking-specific surface_z logic (stack on top of existing object, not table).
- **Narrow containers (baskets with walls)** - need yaw that avoids wall collision; release height above
    basket rim.

---

## Per-Object Grasp Registry (grows per chunk)

Discovered per-object grasp parameters. Coordinator adds entries from `findings.md`.

| Object | z_offset | yaw | Notes | First Seen In |
|---|---|---|---|---|

<!-- PLACEHOLDER: Registry starts empty. Populated by coordinator after each chunk. -->
<!-- Format: object name | z_offset from OBB top | yaw_deg if non-zero | why | task that validated it -->

---

## Anti-Patterns

- `set_joint_qpos` to "teleport" the gripper into grasp pose - forbidden.
- Hardcoded grasp positions from task BDDL - forbidden (reading .bddl is banned).
- Lifting to fixed absolute z (e.g., z=0.9) - use `grasp_pos[2] + 0.15` to be scene-relative.
- Skipping the pre-grasp `z_approach=0.15` - causes gripper collisions with neighbors.
- Single-try `select_top_down_grasp` without `argmax` fallback - some scenes have no valid top-down grasp.

---

## Debugging Grasp Failures

| Symptom | Likely Cause | Check |
|---|---|---|
| Gripper closes on air | z too high (didn't descend to surface) | Print `grasp_pos[2]` vs. `surface_z`;
    use OBB `extent[2]/2` below top |
| Grasp succeeds but drops during lift | Gripper half-closed on slippery/elongated object | Try
    perpendicular yaw (90 degrees) or fingers-along-long-axis approach |
| `plan_grasp` returns empty | Mask too small or below grasp threshold | Dilate mask; log `mask.sum()` -
    should be >200 px |
| `solve_ik` returns None | Pose out of workspace | Lower z by 5cm, re-check x/y reachable from robot base
     |
```

localize initial SKILL.md

```
---
name: localize
description: Object localization via SAM3 - prompting strategies, multi-prompt fallback, 3D centroid
    extraction, disambiguation, per-object prompt registry. Grows through experiment.
---

# Localize - SAM3 Prompting & Object Localization

> **Purpose:** This skill tracks working SAM3 prompts per object + the standard localization helpers.
> **Ownership:** Coordinator adds entries to the prompt registry after reviewing subagent findings.
    Subagents do NOT edit this file.
---

## Standard Localization Helper

```python
import numpy as np

def localize_object(rgb, depth, K, E, prompts):
    """Try prompts in order, return (center, pts, mask) for first hit with >=10 points."""
    depth_img = depth[:, :, 0] if len(depth.shape) == 3 else depth
    if isinstance(prompts, str):
```
```

```
        prompts = [prompts]
    for prompt in prompts:
        masks = segment_sam3_text_prompt(rgb, prompt)
        if not masks:
            continue
        best = max(masks, key=lambda d: d["score"])
        mask = best["mask"].astype(np.uint8)
        pts = mask_to_world_points(mask, depth_img, K, E)
        if pts is None or len(pts) < 10:
            continue
        center = get_oriented_bounding_box_from_3d_points(pts)["center"]
        return center, pts, mask
    return None, None, None
```

Usage:
```python
obs = get_observation()
cam = obs["agentview"]
center, pts, mask = localize_object(
    cam["images"]["rgb"], cam["images"]["depth"],
    cam["intrinsics"], cam["pose_mat"],
    ["<specific prompt>", "<fallback prompt>"]
)
if center is None: raise RuntimeError("Object not found")
```

Why OBB center over mean: `get_oriented_bounding_box_from_3d_points` gives a more robust center for elongated/occluded objects than `pts.mean(axis=0)`.

Fallback: Molmo Pixel Grounding

When SAM3 returns no masks even for sensible prompts, try Molmo to get a pixel, then convert with SAM3 point prompt:

```python
def localize_via_molmo(rgb, depth, K, E, phrase):
    px = point_prompt_molmo(rgb, phrase)  # (x, y) or None
    if px is None: return None, None, None
    masks = segment_sam3_point_prompt(rgb, [px])
    if not masks: return None, None, None
    mask = masks[0]["mask"].astype(np.uint8)
    depth_img = depth[:, :, 0] if len(depth.shape) == 3 else depth
    pts = mask_to_world_points(mask, depth_img, K, E)
    if pts is None or len(pts) < 10: return None, None, None
    return get_oriented_bounding_box_from_3d_points(pts)["center"], pts, mask
```

Disambiguation: Two Similar Objects in Scene

When a scene contains two visually similar objects, `max(score)` often picks the wrong one. Use **bbox pixel area** or **bbox y-position** (upper = farther):

```python
def select_by_smallest_bbox(masks, max_cy=None, min_area=50):
    """Pick mask with smallest bbox pixel area (most compact matching shape).
    max_cy: if set, only accept masks with bbox center-y below this row."""
    ... (implementation omitted)
```

Key: `area_pct` in SAM3 result dicts clamps to 99.00 for all masks - compute area from `box` directly.

```
## Disambiguation: By 3D Position (when two objects share prompts)

‘‘‘python
# Example: "black bowl" returns both bowls, want the one at the front
    ... (implementation omitted)
‘‘‘

---

## Prompt Registry (grows per chunk)

Discovered working prompts, indexed by object. Coordinator appends entries after each chunk review.

**Format per row:** object | priority-ordered prompts | which task validated it | notes.

| Object | Working Prompts | First Seen In | Notes |
|---|---|---|---|

<!-- PLACEHOLDER: Registry starts empty. Populated by coordinator after each chunk. -->
<!-- Format: object name | prompts in priority order (quoted) | first task that validated | disambiguation
      notes if needed -->

---

## How to Add a New Prompt (Coordinator Only)

When a subagent’s ‘findings.md‘ "Generalizable Patterns" section mentions a new working prompt:

1. **Verify generalizability** - would this prompt plausibly work for >=2 task types? If it’s ultra-
     specific to one task, consider keeping it inline in ‘task_code.py‘ instead of here.
2. **Append to registry** - one table row, priority-ordered list of prompts.
3. **Add a disambiguation note** only if the task exposed a specific confusion (e.g., two bowls in frame).
4. **Keep examples short** - a full code block per prompt is overkill. The helper pattern stays at the top
      of this file; the registry is just prompt strings.

---

## Anti-Patterns (Do NOT Do These)

- ‘body_xpos‘ for object localization - forbidden. If you’re tempted, the perception is wrong; fix the
     prompt.
- Hardcoded pixel coordinates - always derive from SAM3 or Molmo.
- Caching object positions across seeds - re-localize every episode; objects are randomized.
- Using ‘segment_sam3_text_prompt‘ mask dict without ‘.astype(np.uint8)‘ conversion - silent failures
     downstream.
```

manipulation initial SKILL.md

```
---
name: manipulation
description: Non-pick-and-place manipulation patterns - drawer opening/closing, knob/switch turning,
     pushing, articulated-object interactions. Grows with validated parameters per object type.
---

# Manipulation - Articulated Objects & Non-Prehensile Actions

> **Purpose:** Tracks validated parameters for articulated manipulation. Approaches are discovered through
      experiment and recorded in the registry below.
> **Ownership:** Coordinator adds validated parameters per object-type. Subagents do not edit.
---

## Examples of Task Types Requiring This Skill

- **Drawer open/close** - pull/push a drawer by its handle
- **Knob/switch turn** - rotate a stove knob or flip a switch to trigger a state change
- **Hinged door** - open a microwave, cabinet door, or similar hinged panel
```

````
- **Non-prehensile push** - slide an object to a target without grasping it

These require non-top-down approach angles, arc motions, or contact-without-grasp patterns.

---

## Parameter Registry (grows per chunk)

Validated parameters per object / action. Coordinator adds after verifying from `findings.md`.

| Object / Action | Parameters | First Seen In | Notes |
|---|---|---|---|

<!-- PLACEHOLDER: Registry starts empty. Populated by coordinator after each chunk. -->
<!-- Format: object/action | key parameters (distances, angles, offsets, approach direction) | first task | notes on what failed before -->

---

## Anti-Patterns

- `set_joint_qpos` to force a drawer/door open - forbidden. Must use physical manipulation.
- Reading BDDL to get pull distances or knob angles - forbidden. Discover via OBB or trial-and-error.
- Forgetting to retreat after manipulation - arm stays inside drawer/microwave, blocks next perception.
````

transport initial SKILL.md

````
---
name: transport
description: Motion patterns for moving the end-effector between poses - multi-step waypoints, safe transit, interpolated Cartesian moves, collision avoidance. Grows with discovered patterns.
---

# Transport - End-Effector Motion Between Poses

> **Purpose:** Transit patterns after a grasp is achieved and before a place/release. Keeps the object above clutter, avoids scene collisions, handles narrow workspaces.
> **Ownership:** Coordinator adds validated waypoint patterns. Subagents do not edit.
---

## Basic Lift-Transit-Descend

```python
# Precondition: object grasped (close_gripper done), grasp_pos known, quat known
# Target: tgt_center (3D world coordinate), surface_z (target surface height)

# Waypoint 1: lift straight up (avoid dragging)
lift_z = grasp_pos[2] + 0.15
joints = solve_ik([grasp_pos[0], grasp_pos[1], lift_z], quat.tolist())
if joints is not None: move_to_joints(joints)

# Waypoint 2: move laterally at lift height (avoid hitting other objects on surface)
joints = solve_ik([tgt_center[0], tgt_center[1], lift_z], quat.tolist())
if joints is not None: move_to_joints(joints)

# Waypoint 3: descend to release height
release_z = surface_z + 0.05
joints = solve_ik([tgt_center[0], tgt_center[1], release_z], quat.tolist())
if joints is not None: move_to_joints(joints)
```

**Why 3 waypoints, not direct goto_pose:** a direct move from grasp to release tends to take a diagonal path that clips tall objects, basket walls, or the drawer opening. Three cardinal waypoints (up → over → down) keeps the object in free air during transit.

---
````

```
## Home Reset Between Subtasks

For long-horizon tasks (LIBERO-Long-Pro), reset the arm between subtasks to a known-safe config:

```python
def safe_reset_between_subtasks():
    open_gripper()
    goto_home_joint_position()
    for _ in range(3): get_observation()  # let physics settle
```

When to call: after releasing subtask-1's object, before starting subtask-2's localization. Prevents
    subtask-2 from trying to perceive through the arm in frame.

Far-Workspace Safety: `solve_ik` over `goto_pose`

`goto_pose` is unreliable in the far workspace (e.g., back of counter, far side of table). Use `solve_ik`
    + `move_to_joints` for these transits:

```python
# UNRELIABLE (far workspace):
# goto_pose(far_target_pos, quat)

# RELIABLE:
joints = solve_ik(far_target_pos.tolist(), quat.tolist())
if joints is None:
    # Try slightly closer z or x
    fallback = far_target_pos.copy(); fallback[2] -= 0.02
    joints = solve_ik(fallback.tolist(), quat.tolist())
if joints is not None:
    move_to_joints(joints)
else:
    raise RuntimeError(f"IK infeasible at {far_target_pos}")
```

Waypoint Registry (grows per chunk)

Validated multi-step transit patterns for scenes where direct motion fails.

Scenario	Pattern	First Seen In	Notes

<!-- PLACEHOLDER: Registry starts empty. Populated by coordinator after each chunk. -->
<!-- Format: scenario description | waypoint pattern used | first task | why default 3-waypoint failed -->

Anti-Patterns

- Single `goto_pose` across the whole scene - clips clutter. Use waypoints.
- Lift waypoint at same z as grasp_pos - drags the object sideways.
- Descending straight into a container without a rim-clearance check - grasp collides with walls.
- Forgetting `for _ in range(3): get_observation()` after `open_gripper()` - object doesn't settle; next
    observation is stale.
- Using `goto_pose` in far workspace where `solve_ik`+`move_to_joints` is required.
```